



Department of Public Health and Environment

Air Quality Control Commission

REGULATION NUMBER 24 CONTROL OF EMISSIONS FROM VOLATILE ORGANIC COMPOUNDS AND PETROLEUM LIQUIDS STORAGE AND PETROLEUM PROCESSING AND REFINING

5 CCR 1001-28

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

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Pursuant to Colorado Revised Statutes Section 24-4-103 (12.5), materials incorporated by reference are available for public inspection during normal business hours, or copies may be obtained at a reasonable cost from the Air Quality Control Commission (the Commission), 4300 Cherry Creek Drive South, Denver, Colorado 80246-1530. The material incorporated by reference is also available through the United States Government Printing Office, online at www.govinfo.gov. Materials incorporated by reference are those editions in existence as of the date indicated and do not include any later amendments.

PART A Applicability and General Provisions

I. Applicability

I.A.

I.A.1. The provisions of this regulation shall apply as follows:

I.A.1.a. All provisions of this regulation apply to the Denver 1-hour ozone attainment/maintenance area, to any nonattainment area for the 1-hour ozone standard, to the 8-hour Ozone Control Area, and to northern Weld County.

I.A.1.b. (State Only) All provisions of this regulation apply to any ozone nonattainment area, which includes areas designated nonattainment for either the 1-hour or 8-hour ozone standard, unless otherwise specified in Section I.A.1.c. Colorado's ozone nonattainment or attainment maintenance area maps and chronologies of attainment status are identified in Appendix A of this regulation.

I.A.1.c. The provisions of Part B, Sections III., IV.B.1. and IV.B.2., and V.C. apply statewide. The provisions marked by (State Only) are not federally enforceable, unless otherwise identified.

I.A.2. REPEALED

I.A.3. REPEALED

I.B. Sources

I.B.1. New Sources

I.B.1.a. New sources, defined as any sources located in the 1-hour ozone nonattainment area or attainment maintenance area, which either (1) submit a complete permit application on or after October 30, 1989, or (2) if no permit is required, commence operation on or

after October 30, 1989, must comply with the provisions of this regulation upon commencement of operation.

I.B.1.b. (State Only) New sources are any sources which commenced construction on or after the date on which the area is first designated as being in nonattainment for ozone and are located in that area, or, if located in the 1-hour ozone nonattainment or attainment maintenance area, by October 30, 1989. New sources shall comply with the requirements of this regulation by whichever date comes later:

I.B.1.b.(i) (State Only) October 30, 1989, if they are located in what was previously designated as a 1-hour ozone nonattainment or attainment maintenance area;

I.B.1.b.(ii) (State Only) February 1, 2009, if they are located in an 8-Hour Ozone Control Area and outside of the 1-hour ozone nonattainment or attainment maintenance area; or

I.B.1.b.(iii) (State Only) Upon commencement of operation, if located within an ozone nonattainment or attainment maintenance area.

I.B.2. Existing Sources

I.B.2.a. Existing sources are

I.B.2.a.(i) (1) those sources for which a complete permit application was submitted prior to October 30, 1989, or (2) those sources, which commenced operation prior to October 30, 1989, if they are located in the 1-hour ozone nonattainment or attainment maintenance area.

I.B.2.a.(ii) Those sources that submitted a complete permit application or that commenced operation prior to February 1, 2009, if they are located in the 8-Hour Ozone Control Area and outside of the 1-hour ozone nonattainment or attainment maintenance area.

I.B.2.a.(iii) Those sources located in northern Weld County that submitted a complete application or that commenced operation prior to February 14, 2023.

I.B.2.b. (State Only) Existing sources located in the 1-hour ozone nonattainment or attainment maintenance area are those sources which commenced construction prior to the date on which the area is first designated as being in nonattainment for ozone and are located in that area, or, if located in the 1-hour ozone

nonattainment or attainment maintenance area, by October 30, 1989.

I.B.2.c. Existing sources located in the 1-hour ozone nonattainment or attainment maintenance area shall not be required to comply with requirements of this regulation until on and after October 30, 1991. All existing sources shall comply with the requirements set forth in Exhibit A until October 30, 1991.

I.B.2.d. Existing sources shall be required to comply with requirements of this regulation by the following dates.

I.B.2.d.(i) (State Only) October 30, 1989, if they are located in what was previously designated as a 1-hour ozone nonattainment or attainment maintenance area;

I.B.2.d.(ii) (State Only) February 1, 2009, if they are located in an 8-hour Ozone Control Area and outside of the Denver 1-hour ozone nonattainment or attainment maintenance area;

I.B.2.d.(iii) February 14, 2023, if located in northern Weld County and subject to requirements in Part B, Sections III., IV.B.1., IV.B.2., or V.C.

I.B.2.d.(iv) December 31, 2024, if located in northern Weld County and not subject to requirements in Part B, Sections III., IV.B.1., IV.B.2., or V.C.; or

I.B.2.d.(v) (State Only) Three years after the date on which the area is first designated as being in nonattainment for ozone, if located within that ozone nonattainment or attainment maintenance area.

I.B.2.e. On and after October 30, 1991, all existing sources located in the 1-hour ozone nonattainment or attainment maintenance area shall comply with the requirements of this regulation, and Exhibit A shall no longer be applicable.

I.B.2.f. Repealed.

I.B.2.g. Repealed.

I.C. Once a source subject to this regulation exceeds an applicable threshold limit, the requirements of this regulation are irrevocably effective unless the source obtains a federally enforceable permit limiting emissions to levels below the threshold limit by restricting production capacity or hours of operation.

- I.D. The owner or operator of a source not required to obtain a permit by provisions of law other than this section may apply for and shall be required to accept a permit as a condition of avoiding RACT requirements. Such permits shall contain only those conditions necessary to ensure the enforcement of the production capacity or hours of operation.

II. General Provisions

II.A. Definitions

- II.A.1. “8-Hour Ozone Control Area” means the Counties of Adams, Arapahoe, Boulder (includes part of Rocky Mountain National Park), Douglas, and Jefferson; the Cities and Counties of Denver and Broomfield; and the following portions of the Counties of Larimer and Weld:
- II.A.1.a. For Larimer County (includes part of Rocky Mountain National Park), that portion of the county that lies south of a line described as follows: Beginning at a point on Larimer County’s eastern boundary and Weld County’s western boundary intersected by 40 degrees, 42 minutes, and 47.1 seconds north latitude, proceed west to a point defined by the intersection of 40 degrees, 42 minutes, 47.1 seconds north latitude and 105 degrees, 29 minutes, and 40.0 seconds west longitude, thence proceed south on 105 degrees, 29 minutes, 40.0 seconds west longitude to the intersection with 40 degrees, 33 minutes and 17.4 seconds north latitude, thence proceed west on 40 degrees, 33 minutes, 17.4 seconds north latitude until this line intersects Larimer County’s western boundary and Grand County’s eastern boundary.
- II.A.1.b. For Weld County, that portion of the county that lies south of a line described as follows: Beginning at a point on Weld County’s eastern boundary and Logan County’s western boundary intersected by 40 degrees, 42 minutes, 47.1 seconds north latitude, proceed west on 40 degrees, 42 minutes, 47.1 seconds north latitude until this line intersects Weld County’s western boundary and Larimer County’s eastern boundary.
- II.A.2. “Denver 1-Hour Ozone Attainment/Maintenance Area” means the Counties of Jefferson and Douglas, the Cities and Counties of Denver and Broomfield, Boulder County (excluding Rocky Mountain National Park), Adams County west of Kiowa Creek, and Arapahoe County west of Kiowa Creek.
- II.A.3. “Capture System” means the equipment used to contain, capture, or transport a pollutant to a control device.
- II.A.4. “Capture System Efficiency (vapor gathering system efficiency)” means the percent by weight of VOC emitted by an operation subject to this

regulation, which is captured by the capture system and sent to the control device; i.e., $(\text{mass flow of VOC captured})/(\text{mass flow of VOC emitted by the operation}) \times 100\%$.

- II.A.5. "Carbon Adsorption System" means a device containing adsorbent material, an inlet and outlet for exhaust gases and a system to regenerate the saturated adsorbent.
- II.A.6. "Condenser" means any heat transfer device used to liquefy vapors by removing their latent heats of vaporization. Such devices include, but are not limited to, shell and tube, coil, surface, or contact condensers.
- II.A.7. "Control Device" means a carbon adsorber, refrigeration system, condenser, flare, firebox or other device, which will reduce the concentration of VOC in a gas stream by adsorption, combustion, condensation, or other means of removal.
- II.A.8. "Control Device Efficiency" means the percent removal by weight of VOC by a control device; i.e., $(\text{mass flow of VOC into control device} - \text{mass flow of VOC out of control device})/(\text{mass flow of VOC into control device}) \times 100\%$.
- II.A.9. "Gasoline" means a petroleum distillate having a Reid vapor pressure between 208 and 1040 torr (4-20 psi), which is used as fuel for internal combustion engines.
- II.A.10. "Highly Volatile Organic Compound" is defined as a Volatile Organic Compound or mixture of such compounds with a true vapor pressure in excess of 570 torr (11 psia) at 20 C.
- II.A.11. "Organic Material" means a chemical compound of carbon, excluding carbon monoxide, carbon dioxide, carbonic acid, metallic carbides or carbonates, and ammonium carbonate.
- II.A.12. (State Only) "Ozone Nonattainment Area" means any area designated as not in attainment with the ozone National Ambient Air Quality Standard as determined by the Environmental Protection Agency.
- II.A.13. "Northern Weld County" means the portion of the county that does not lie south of a line described as follows: Beginning at a point on Weld County's eastern boundary and Logan County's western boundary intersected by 40 degrees, 42 minutes, 47.1 seconds north latitude, proceed west on 40 degrees, 42 minutes, 47.1 seconds north latitude until this line intersects Weld County's western boundary and Larimer County's eastern boundary.
- II.A.14. "Petroleum Refinery" means any facility engaged in producing gasoline, aromatics, kerosene, distillate fuel oils, residual fuel oils,

lubricants, asphalt, or other products through distillation of petroleum or through redistillation, cracking, rearrangement or reforming of unfinished petroleum derivatives.

- II.A.15. "Reid Vapor Pressure" means the absolute vapor pressure of volatile crude oil and volatile nonviscous petroleum liquids except liquefied petroleum gases as determined by the American Society for Testing and Materials, Part 17, 1973, D-323-72 (Reapproved 1977).
- II.A.16. "True Vapor Pressure" means the equilibrium partial pressure exerted by petroleum (or other) liquid. This may be determined by the methods described in American Petroleum Institute Bulletin 2517, "Evaporation Loss from Floating Roof Tanks," 1962.
- II.A.17. "Vapor Recovery System" means a system that prevents release to the atmosphere of organic compounds emitted during the operation of any transfer, storage, or processing equipment.
- II.A.18. "Volatile Organic Compound (VOC)" means any compound of carbon, excluding carbon monoxide, carbon dioxide, carbonic acid, metallic carbides or carbonates, and ammonium carbonate, which participates in atmospheric photochemical reactions, except those listed in Section II.B. as having negligible photochemical reactivity. VOC may be measured by a reference method, an equivalent method, an alternative method, or by procedures specified under 40 CFR Part 60 (July 1, 2022). A reference method, an equivalent method, or an alternative method, however, may also measure nonreactive organic compounds. In such cases, an owner or operator may exclude the compounds listed in Section II.B. when determining compliance with a standard if the amount of such compounds is accurately quantified, and such exclusion is approved by the Division. As a precondition to excluding such compounds as VOC, or at any time thereafter, the Division may require an owner or operator to provide monitoring or testing methods and results demonstrating, to the satisfaction of the Division, the amount of negligible-reactive compounds in the source's emissions.

II.B. Exemptions

Emissions of the organic compounds listed as having negligible photochemical reactivity in the common provisions definition of Negligibly Reactive Volatile Organic Compound are exempt from the provisions of this regulation.

II.C. General Emission Limitation

- II.C.1. Existing Sources located in the 1-hour ozone nonattainment or attainment maintenance area or the 8-Hour Ozone Control Area

- II.C.1.a. All existing sources shall comply with the requirements set forth in this regulation.
 - II.C.1.a.(i) Existing sources of VOC which are not subject to specific emission limitations set forth in this regulation, and which have the potential to emit 100 tons per year or more of VOC, shall utilize Reasonably Available Control Technology (RACT).
 - II.C.1.a.(ii) The potential to emit of such sources shall be based on design capacity or maximum production rate, whichever is greater, 8,760 hours/year operation, and before add-on controls.
 - II.C.1.a.(iii) Owners or operators of such sources with potential emissions of 100 tons per year or more, but with actual emissions less than 100 tons per year may obtain a federally enforceable permit limiting emissions to actual rates by restricting production capacity or hours of operation, thus avoiding RACT requirements. The owner or operator of a source not required to obtain a permit by provisions of law other than this section may apply for and shall be required to accept a permit as a condition of avoiding RACT requirements. Such permits shall contain only those conditions necessary to ensure the enforcement of the production capacity or hours of operation.
 - II.C.1.a.(iv) Such sources with potential emissions of 100 tons per year or more but with actual emissions of less than 50 tons per year, on a rolling 12-month total, may avoid RACT and permit requirements if the following requirements are met:
 - II.C.1.a.(iv)(A) The owner or operator shall submit revised Air Pollutant Emission Notices (APENs) by April 1 of each year, which demonstrate that the 50 tons per year threshold has not been exceeded.
 - II.C.1.a.(iv)(B) The owner or operator shall maintain records on site which include monthly VOC use and monthly VOC emissions. The records shall include calculation of total emissions for each rolling 12-month period. The records shall be made available to the Division for inspection upon request.
 - II.C.1.a.(v) (State Only) Existing sources that are modified – undergo any physical change, or changed in the method of operation of a stationary source which increase VOC or NOx

emissions – on or after March 30, 2008, shall utilize RACT control technologies pursuant to Regulation Number 7, Number 24, Number 25, and Number 26 and Regulation Number 3, Part B, Section III.D.2. upon recommencing operation.

II.C.1.b. Provided however, that no existing source of VOC emissions employing emission controls on or within the six-month period preceding the effective date of this regulation may reduce its level of control of VOC emissions below that level of control actually achieved, even though such source may otherwise be subject to less stringent control requirements, except that no existing source shall be required to control emissions to an extent greater than that level of control which RACT would achieve.

II.C.1.c. (State Only) Existing sources with potential emissions equal to or greater than 100 tons per year of volatile organic compound emissions shall submit a permit modification application that includes a revised APEN (or APENs) and a RACT analysis, to the Division, as follows:

II.C.1.c.(i) (State Only) By October 30, 1991 if located in what was previously designated as the Denver 1-hour ozone nonattainment or attainment maintenance area; or

II.C.1.c.(ii) (State Only) By April 30, 2009 or within one year after the date on which the area is first designated as being in nonattainment for ozone, whichever comes later, if they are located in the 8-hour Ozone Control Area and outside of the Denver 1-hour ozone nonattainment or attainment maintenance area.

II.C.1.d. (State Only) Existing sources shall utilize RACT pursuant to Regulation Number 7, Number 24, Number 25, and Number 26 and Regulation Number 3, Part B, Section III.D.2. by whichever date comes later:

II.C.1.d.(i) (State Only) October 30, 1991, if they are located in what was previously designated as the Denver 1-hour ozone nonattainment or attainment maintenance area;

II.C.1.d.(ii) (State Only) November 21, 2011, if they are located in the 8-hour Ozone Control Area, and outside of the Denver 1-hour ozone nonattainment or attainment maintenance area;

II.C.1.d.(iii) (State Only) Three years after the date on which the area is first designated as being in nonattainment for ozone; or

II.C.1.d.(iv) (State Only) Two years after Division determination of case-by-case RACT pursuant to this Section II.C.1. The Division shall be deemed to have approved the RACT analysis for purposes of this Section II.C.1.d.(iv) if it does not object after eighteen months from having received a complete permit application.

II.C.2. New Sources

All new sources shall utilize controls representing RACT, pursuant to Regulation Number 7, Number 24, Number 25, and Number 26 and Regulation Number 3, Part B, Section III.D., upon commencement of operation.

II.D. Alternative Control Plans and Test Methods

II.D.1. Sources subject to specific requirements of this regulation shall submit for approval as a revision to the State Implementation Plan:

II.D.1.a. Any alternative emission control plan or compliance method other than control options specifically allowed in the applicable regulation. Such alternative control plans shall provide control equal to or greater than the emission control or reduction required by the regulation, unless the source contends that the control level required by the regulation does not represent RACT for their specific source.

II.D.1.b. Any alternative test method or procedure not specifically allowed in the applicable regulation.

II.D.2. No alternative submitted pursuant to this Section II.D. is effective until the alternative is approved as a revision to the State Implementation Plan.

II.E. REPEALED

II.F. Provisions for Specific Processes

II.F.1. The Gates Rubber Company Provision – REPEALED

Appendix A Colorado Ozone Nonattainment or Attainment Maintenance Areas

I. Chronology of Attainment Status

Denver Metropolitan Area Only

1978 Denver 1-hour Ozone Nonattainment Area designation first becomes effective in 7-county Denver Metropolitan Area

10/11/01 Denver 1-hour Ozone Attainment Maintenance Area designation replaces non-attainment designation and becomes effective in 7-county Denver Metropolitan Area

9/2/05 1-hour Ozone National Ambient Air Quality Standard is Revoked in Colorado except for the Denver 1-hour Ozone Attainment Maintenance Area.

Denver Metropolitan Area and North Front Range

10/11/01 1-hour attainment maintenance area replaces non-attainment designation for the Denver Metro Area/North Front Range Area

4/15/04 EPA designates the Denver Metro Area/North Front Range region as an 8-hour ozone non-attainment area, designation deferred due to the implementation of the Early Action Compact

11/20/07 Denver 8-hour ozone non-attainment designation (1997 NAAQS) becomes effective in 9 county Denver Metropolitan Area

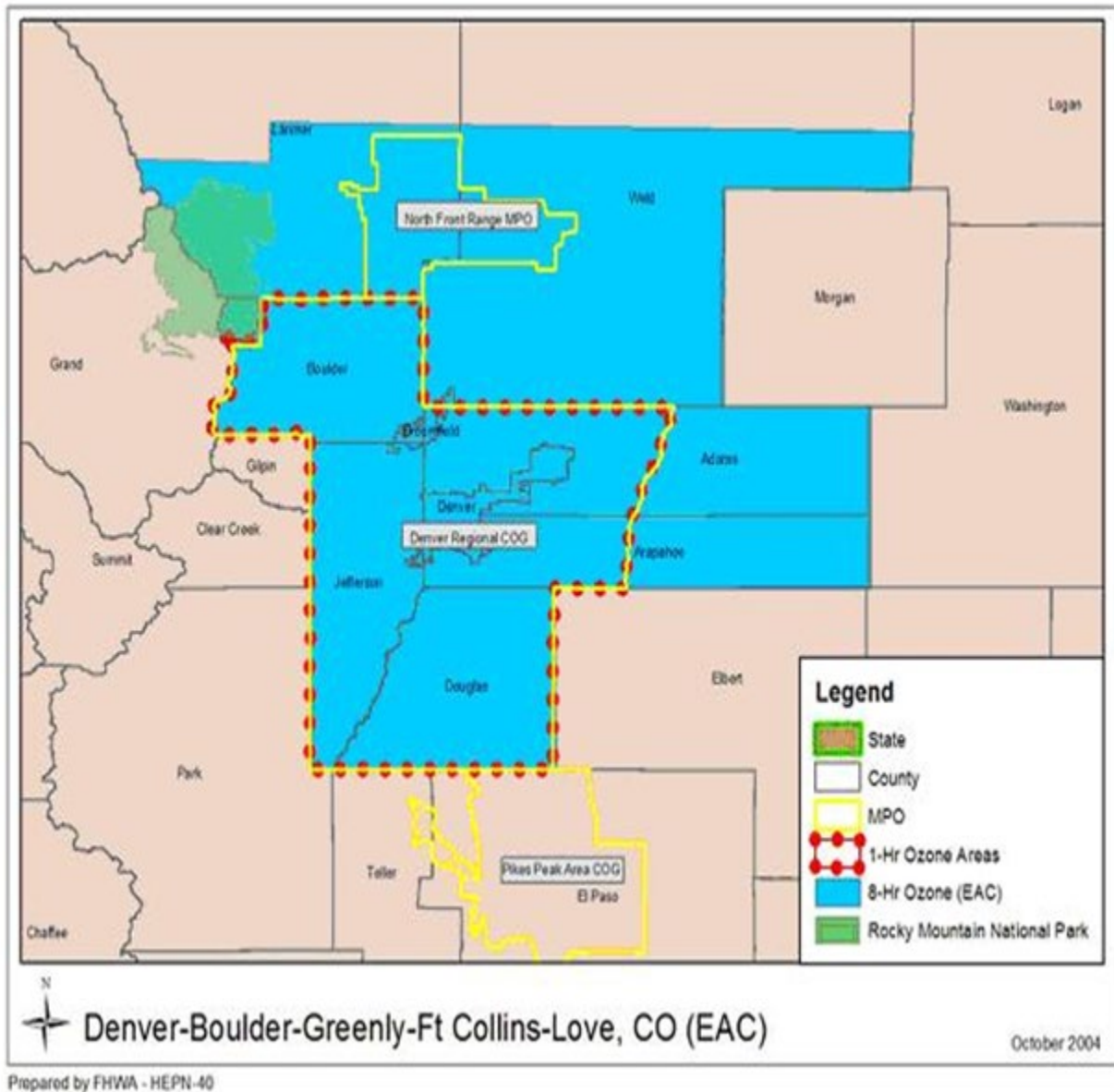
7/20/2012 Denver 8-hour ozone non-attainment designation (2008 NAAQS) becomes effective in 9 county Denver Metropolitan Area

8/3/2018 Denver 8-hour ozone nonattainment designation (2015 NAAQS) becomes effective in 9 county Denver Metropolitan Area

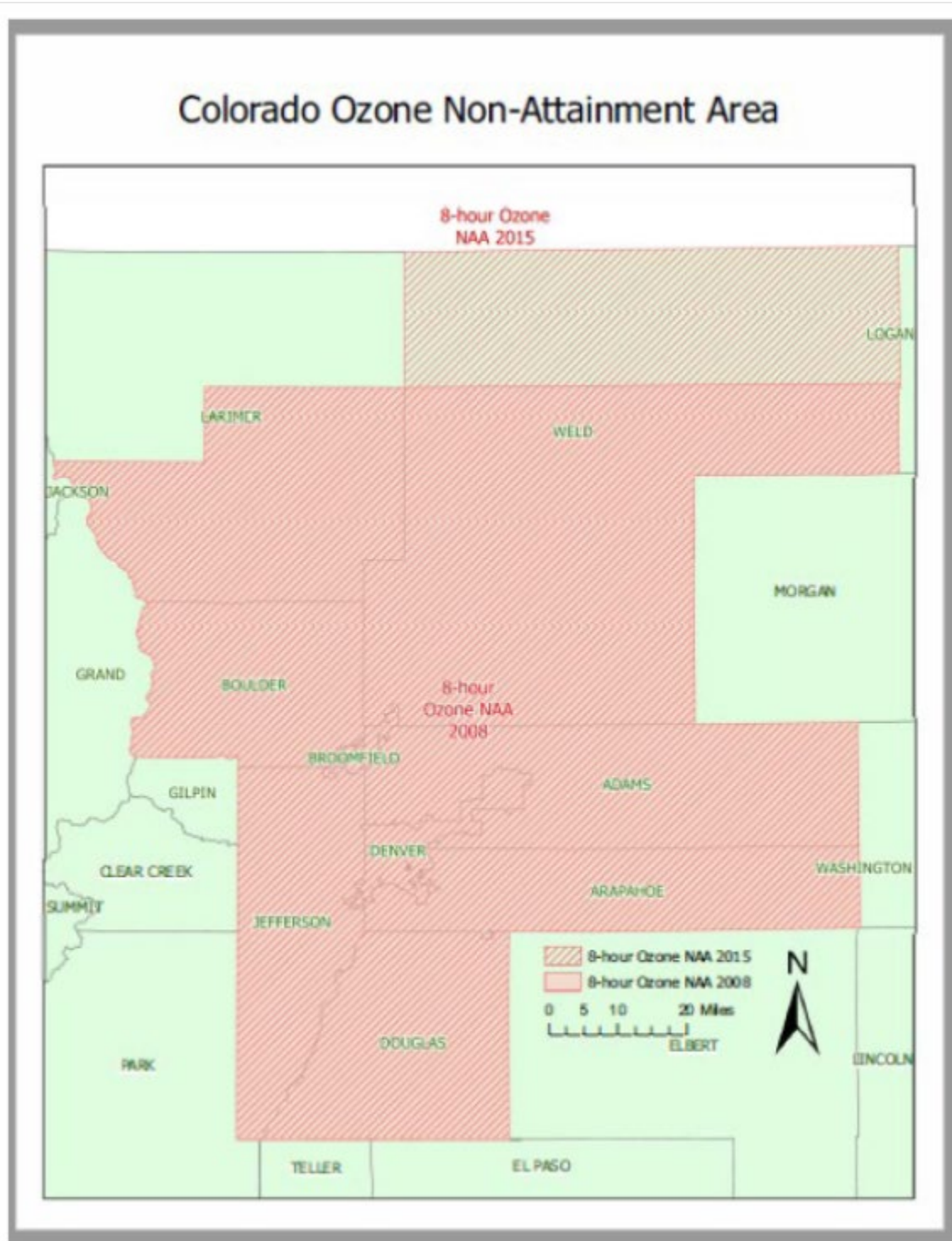
12/31/2021 EPA modification of the 9 county Denver Metropolitan Area 8-hour ozone nonattainment designation (2015 NAAQS) to include the portion of northern Weld County defined in Part A

II. Maps

Denver Metropolitan Area and North Front Range (2008 Ozone NAAQS)



**Denver Metropolitan Area and North Front Range and northern Weld County
(2015 ozone NAAQS)**



PART B Storage, Transfer, and Disposal of Volatile Organic Compounds and Petroleum Liquids and Petroleum Processing and Refining

I. General Requirements for Storage and Transfer of Volatile Organic Compounds

I.A. Maintenance and Operation of Storage Tanks and Related Equipment

All storage tank gauging devices, anti-rotation devices, accesses, seals, hatches, roof drainage systems, support structures, and pressure relief valves shall be maintained and operated to prevent detectable vapor loss except when opened, actuated, or used for necessary and proper activities (e.g. maintenance). Such opening, actuation, or use shall be limited so as to minimize vapor loss.

Detectable vapor loss shall be determined visually, by touch, by presence of odor, or using a portable hydrocarbon analyzer. When an analyzer is used, detectable vapor loss means a VOC concentration exceeding 10,000 ppm. Testing and monitoring shall be conducted as in Part B, Section VI.C.3.

I.B. Transfer (excluding Petroleum Liquids)

Except as otherwise provided in this regulation, all volatile organic compounds transferred to any tank, container, or vehicle compartment with a capacity exceeding 212 liters (56 gallons), shall be transferred using submerged or bottom filling equipment. For top loading, the fill tube shall reach within six inches of the bottom of the tank compartment. For bottom-fill operations, the inlet shall be flush with the tank bottom.

I.C. Beer production and associated beer container storage and transfer operations involving volatile organic compounds with a true vapor pressure of less than 1.5 psia actual conditions are exempt from the provisions of Section I.B.

II. Storage of Highly Volatile Organic Compounds

II.A. Highly volatile organic compounds shall be stored:

II.A.1. In a pressure tank which is at all times capable of maintaining working pressures sufficient to prevent vapor loss to the ambient air; or

II.A.2. With methods and/or equipment approved by the Division in writing pursuant to the request of the person owning or operating the storage facility.

II.B. Vapor loss shall be determined visually, by presence of frost or condensation at the point of leakage, or using a portable hydrocarbon analyzer. When an analyzer is used, vapor loss means a VOC concentration exceeding 10,000 ppm and testing and monitoring procedures shall be conducted as in Part B, Section VI.C.3.

III. Disposal of Volatile Organic Compounds

- III.A. No person shall dispose of volatile organic compounds by evaporation or spillage unless RACT is utilized.
- III.B. No owner or operator of a bulk gasoline terminal, bulk gasoline plant, or gasoline dispensing facility as defined in Part B, Sections IV.C.2., IV.C.3. and VII.A.3., shall permit gasoline to be intentionally spilled, discarded in sewers, stored in open containers, or disposed of in any other manner that would result in evaporation.

IV. Storage and Transfer of Petroleum Liquid

IV.A. General Requirements

- IV.A.1. No person shall build, install, or permit the building or installation of any rotating pump or compressor handling any type of petroleum liquid unless said pump or compressor is equipped with mechanical seals or other equipment of equal efficiency. If reciprocating-type pumps and compressors are used, they shall be equipped with packing glands properly installed, in good working order, and properly maintained so that no detectable emissions occur from the drain recovery systems.

IV.A.2. Definitions

For the purpose of this section, the following definitions apply:

- IV.A.2.a. Repealed.

- IV.A.2.b. "Crude Oil" means a naturally occurring mixture which consists of hydrocarbons, sulfur, nitrogen or oxygen derivatives of hydrocarbons, and which is a liquid at standard conditions.

- IV.A.2.c. "Custody Transfer" means the transfer of produced crude oil and/or condensate, after processing and/or treating in the producing operations, from storage tanks or automatic transfer facilities to pipelines or any other forms of transportation.

- IV.A.2.d. "EFR Tank" means a storage vessel having an external floating roof.

- IV.A.2.e. "External Floating Roof" means a storage vessel cover in an open top tank consisting of a double deck or pontoon single deck which rests upon and is supported by the petroleum liquid being contained and is equipped with a closure seal or seals to close the space between the roof edge and tank wall.

- IV.A.2.f. "Liquid-Mounted Seal" means a primary seal mounted in continuous contact with the contained liquid and which occupies an annular space between the inner tank wall and the perimeter of the floating roof.
- IV.A.2.g. "Petroleum Liquid" means crude oil, condensate and any finished or intermediate product manufactured or extracted in a petroleum refinery.
- IV.A.2.h. "Shoe Seal" means a primary seal employing a metallic band (called a shoe) which is held against the vertical inner-wall of the tank, concentric with the perimeter of the floating roof.
- IV.A.2.i. "Vapor Balance System" means a combination of pipes or hoses that create a closed system between the vapor spaces of an unloading tank and a receiving tank such that vapors displaced from the receiving tank are transferred to the tank being unloaded.
- IV.A.2.j. "Vapor Collection System" means a vapor transport system which uses direct displacement by the gasoline being transferred to force vapors from the vessel being loaded into either a vessel being unloaded or a vapor holding tank.
- IV.A.2.k. "Vapor-Mounted Seal" means a primary seal mounted so there is an annular vapor space underneath the seal. The annular vapor space is bounded by the bottom of the primary seal, the liquid surface, the floating roof, and the tank wall (thus excluding shoe seals).
- IV.A.2.l. "Waxy, Heavy Pour Crude Oil" means a crude oil with a pour point of 10°C (50°F) or higher as determined by the American Society for Testing and Materials Standard D97-66, "Test for Pour Point of Petroleum Oils."

IV.B. Storage of Petroleum Liquid

IV.B.1. Exemptions

- IV.B.1.a. Tanks or other containers used to store the following liquids are exempt from the provisions of Sections IV.B.2. and IV.B.3.:
- IV.B.1.a.(i) Diesel Fuels 1-D, 2-D, and 4-D as defined in ASTM D975-78.
- IV.B.1.a.(ii) Fuel Oils #1, #2, #3, #4, and #5, as defined in ASTM D396-78.

IV.B.1.a.(iii) Gas Turbine Fuels 1-GT through 4-GT as defined in ASTM D2880-78.

IV.B.1.b. The following underground storage facilities are exempt from Section IV.B.2.:

IV.B.1.b.(i) Underground tanks if the annual sum total of the volume of liquid removed from the tank plus the sum of the volume of liquid added to it does not exceed twice the operational volume of the tank (i.e., a maximum of one turnover per year is allowed).

IV.B.1.b.(ii) Subsurface caverns or porous rock reservoirs.

IV.B.1.b.(iii) Horizontal underground tanks storing JP-4 Jet Fuel.

IV.B.2. Storage of petroleum liquid in tanks greater than 151,412 liters (40,000 gallons)

IV.B.2.a. Storage of petroleum liquid in fixed-roof tanks.

IV.B.2.a.(i) The owner or operator of a fixed-roof tank used for storage of petroleum liquids which have a true vapor pressure greater than 33.6 torr (0.65 psia) at 20°C (or, alternatively, a Reid vapor pressure greater than 1.30 pounds - (67.2 torr) but not greater than 570 torr (11.0 psia) at 20°C, and which are stored in any tank or other container of more than 151,412 liters (40,000 gallons) shall ensure that the tank at all times meets the following conditions:

IV.B.2.a.(i)(A) The tank has been equipped with a pontoon-type, or double-deck type, floating roof or an internal floating cover which rests on the surface of the liquid contents and which is equipped with a closure seal or seals to close the space between the edge of the floating roof (or cover) and tank walls; or

IV.B.2.a.(i)(B) The tank has been equipped with a vapor gathering system capable of collecting the petroleum liquid vapors discharged, together with a vapor recovery or disposal system capable of processing such vapors so as to prevent their emission into the atmosphere.

IV.B.2.a.(i)(C) Control devices shall meet the applicable requirements, including recordkeeping, of Regulation Number 25, Part B, Sections I.A.3.a., b., c., and e., and I.A.8.a. and b.

IV.B.2.a.(i)(D) The applicable EPA reference methods 1 through 4, and 25, of 40 CFR Part 60 (March 23, 2021) shall be used to determine the efficiency of control devices.

IV.B.2.a.(i)(E) The owner or operator shall maintain records for at least two years of the type, average monthly storage temperature, and true vapor pressure of all petroleum liquids stored in tanks not equipped with an internal floating roof or cover or other control pursuant to Regulation Number 24, Sections IV.B.2.a.(i)(A) or IV.B.2.a.(i)(B) or Part A, Section II.D.

IV.B.2.a.(ii) No owner or operator of a fixed-roof tank equipped with an internal floating roof or cover shall permit the use of such tank unless:

IV.B.2.a.(ii)(A) The tank is maintained such that there are no visible holes, tears, or other openings in the seal or any seal fabric or materials; and

IV.B.2.a.(ii)(B) All openings, except stub drains, are equipped with covers, lids, or seals such that:

IV.B.2.a.(ii)(B)(1) The cover, lid, or seal is in the closed position at all times except when in actual use;

IV.B.2.a.(ii)(B)(2) Automatic bleeder vents are closed at all times except when the roof is floated off or landed on the roof leg supports; and

IV.B.2.a.(ii)(B)(3) Rim vents, if provided, are set to open when the roof is being floated off the roof leg supports or at the manufacturer's recommended setting.

IV.B.2.a.(iii) The operator of a fixed-roof tank equipped with an internal floating roof shall:

IV.B.2.a.(iii)(A) Perform a routine inspection through the tank roof hatches at least once every six months;

IV.B.2.a.(iii)(A)(1) During the routine inspection, the operator shall measure for detectable vapor loss inside the hatch. Detectable vapor loss

means a VOC concentration exceeding 10,000 ppm, using a portable hydrocarbon analyzer.

IV.B.2.a.(iii)(B) Perform a complete inspection of the cover and seal whenever the tank is out of service, whenever the routine inspection required in Section IV.B.2.a.(iii)(A) reveals detectable vapor loss, and at least once every ten years, and shall notify the Division in writing before such an inspection.

IV.B.2.a.(iii)(C) Ensure during inspections that there are no visible holes, tears, or other openings in the seal or any seal fabric or materials; that the cover is floating uniformly on or above the liquid surface; that there are no visible defects in the surface of the cover or liquid accumulated on the cover; and that the seal is uniformly in place around the circumference of the cover between the cover and the tank wall. If these items are not met, the owner or operator shall repair the items or empty and remove the storage vessel from service within 45 days. If a failure that is detected during inspections required in this section cannot be repaired within 45 days and if the vessel cannot be emptied within 45 days, a 30-day extension may be requested from the Division in writing. Such a request must document that alternative storage capacity is unavailable and specify a schedule of actions the owner or operator will take that will assure that the items will be repaired or the vessel will be emptied as soon as possible;

IV.B.2.a.(iii)(D) Maintain records for at least two years of the results of all inspections.

IV.B.2.b. Above ground storage tanks used for the storage of petroleum liquid shall have all external surfaces coated with a material which has a reflectivity for solar radiation of 0.7 or more. Methods A or B of ASTM E424 shall be used to determine reflectivity. Alternatively, any untinted white paint may be used which is specified by the manufacturer for such use.

This provision shall not apply to written symbols or logograms applied to the external surface of the container for purposes of identification provided such symbols do not cover more than 20% of the exposed top and side surface area of the container or more than 18.6 square meters (200 square feet), whichever is less.

IV.B.2.c. Seals on External Floating Roof Tanks

IV.B.2.c.(i) General Provisions

IV.B.2.c.(i)(A) Applicability

This section applies to all petroleum liquid storage vessels equipped with external floating roofs, having capacities greater than 150,000 liters (40,000 gallons) that are located in ozone nonattainment areas.

IV.B.2.c.(i)(B) Exemptions

IV.B.2.c.(i)(B)(1) Total Exemption

The following categories of EFR tanks are exempt from the requirement of Section IV.B.2.c., except for the applicable recordkeeping requirements of Section IV.B.2.c.(ii)(C).

IV.B.2.c.(i)(B)(1)(a) EFR tanks which store any material whose true vapor pressure as stored never exceeds 67 torr (1.3 psia).

IV.B.2.c.(i)(B)(1)(b) Tanks less than 1,600,000 liters (10,000 barrels) which are used to store crude oil and condensate prior to custody transfer.

IV.B.2.c.(i)(B)(2) Limited Exemptions

The following are exempt from both secondary seal and secondary seal inspection requirements but shall meet the equipment/procedure provisions in Section IV.B.2.c.(ii)(A)(1), the semi-annual inspection provisions of Section IV.B.2.c.(ii)(B), and the record keeping provisions of Section IV.B.2.c.(ii)(C).

IV.B.2.c.(i)(B)(2)(a) Those tanks storing petroleum liquid between 67 and 207 torr (1.3 to 4.0 psia) maximum true vapor pressure (as stored) which are of welded construction and which have one of the following primary seals:

IV.B.2.c.(i)(B)(2)(a)(I) metallic shoe seal

IV.B.2.c.(i)(B)(2)(a)(II) liquid
mounted, resilient seal

IV.B.2.c.(i)(B)(2)(a)(III) liquid
mounted, liquid filled seal

IV.B.2.c.(i)(B)(2)(b) Any tank storing waxy,
heavy-pour crude oil.

IV.B.2.c.(ii) General Requirements

IV.B.2.c.(ii)(A) An operator of an EFR tank storing
petroleum liquids with true vapor pressure (as stored)
above 67 torr (1.3 psia) shall equip the tank as follows
and observe the following procedures:

IV.B.2.c.(ii)(A)(1) Equipment

IV.B.2.c.(ii)(A)(1)(a) Drains: roof drains which
are designed to empty directly into the
stored product shall be provided with
slotted-membrane fabric covers or
equivalent covers which cover at least
90 percent of the area of the opening.

IV.B.2.c.(ii)(A)(1)(b) Openings: except for
automatic bleeder vents, rim space
vents, and leg sleeves, all openings
shall be equipped with:

IV.B.2.c.(ii)(A)(1)(b)(I) Projections
into the tank which remain below
the liquid surface at all times; and

IV.B.2.c.(ii)(A)(1)(b)(II) Covers,
seals, or lids.

IV.B.2.c.(ii)(A)(2) Procedures

IV.B.2.c.(ii)(A)(2)(a) Covers, seals and lids
shall be kept closed except when the
openings are in actual use.

IV.B.2.c.(ii)(A)(2)(b) Automatic bleeder vents
shall be kept closed at all times except
when the roof is floated off or landed on
roof leg supports.

IV.B.2.c.(ii)(A)(2)(c) Rim vents shall be set to open at the manufacturer's recommended setting or, alternatively, only when the roof is being floated off the leg supports.

IV.B.2.c.(ii)(B) Inspections

The operator of an EFR tank subject to this Section IV.B.2.c. shall:

IV.B.2.c.(ii)(B)(1) Perform routine inspections at least once every six months in order to ensure compliance with Section IV.B.2.c.(ii)(B)(2). The inspections shall include a visual inspection of the secondary seal gap if equipped with a secondary seal.

IV.B.2.c.(ii)(B)(2) Ensure that all seal closure devices meet the following requirements:

IV.B.2.c.(ii)(B)(2)(a) There are no visible holes, tears, or other openings in the seal(s) or seal fabric; and

IV.B.2.c.(ii)(B)(2)(b) The seal(s) are intact and uniformly in place around the circumference of the floating roof and the tank wall.

IV.B.2.c.(ii)(C) Records

IV.B.2.c.(ii)(C)(1) Operators shall:

IV.B.2.c.(ii)(C)(1)(a) Maintain records of the average monthly storage temperature, the Reid vapor pressure of the liquid and the type of liquid stored for all EFR tanks lacking secondary seals and receiving petroleum liquids with a true vapor pressure of 1.0 psi (7.0kPa) or greater; and

IV.B.2.c.(ii)(C)(1)(b) Maintain records of the results of the inspections required herein.

IV.B.2.c.(ii)(C)(2) Copies of all records specified by this Section IV.B.2.c.(ii)(C) shall be retained by the operator for a minimum of two years after the date on which the record was made.

IV.B.2.c.(iii) Secondary Seal Requirements

IV.B.2.c.(iii)(A) General

No owner or operator of an EFR tank (storing petroleum liquids) not specifically exempted in Section IV.B.2.c.(i)(B) shall store that petroleum liquid unless such vessel is equipped with a continuous secondary seal extending from the rim of the floating roof to the tank wall (i.e., a rim-mounted secondary seal).

IV.B.2.c.(iii)(B) Vapor-Mounted Seals

For EFR tanks required to have a secondary seal and which have a vapor-mounted primary seal:

IV.B.2.c.(iii)(B)(1) An annual inspection shall be made of the total gap area between the secondary seal and the wall of the tank in accordance with the method in IV.B.2.c.(iii)(B)(3).

IV.B.2.c.(iii)(B)(2) This total gap area shall not exceed 21.2 cm²/meter diameter (1.0 in²/ft. diameter).

IV.B.2.c.(iii)(B)(3) Method to determine gap area:

IV.B.2.c.(iii)(B)(3)(a) Physically measure the length and width of all gaps around the entire circumference of the secondary seal in each place where a 0.32 cm (1/8 in.) uniform diameter probe passes freely (without forcing or binding against the seal) between the seal and the tank wall; and,

IV.B.2.c.(iii)(B)(3)(b) Sum the area of the individual gaps.

IV.B.3. Storage of petroleum liquid in tanks of or less than 151,412 liters (40,000 gallons) capacity.

- IV.B.3.a. Tanks or containers used to store liquids with true vapor pressure at 20°C of less than 78 torr (1.5 psia) or greater than 570 torr (11.0 psia) at 20°C are exempt from the provisions of this Section IV.B.3.
- IV.B.3.b. The owner or operator of storage tanks at a gasoline dispensing facility (service station) or other facility not addressed in Sections IV.C.2. or IV.C.3., which receives and stores petroleum liquid, shall not allow the transfer of petroleum liquid from any delivery vessel into any tank unless the tank is equipped with a submerged fill pipe and all vapors displaced from the storage tank are transferred to the delivery vessel being unloaded using a properly maintained, functioning, and leak-tight vapor collection system, as in accordance with applicable provisions of Appendix B and Section VII., if the tank:
 - IV.B.3.b.(i) Has a rated manufacturer's capacity of 2,082 liters (550 gallons) or more and was installed after November 7, 1973, (except for storage tanks below 550-gallon capacity used exclusively for agricultural use; however, these must have a submerged fill pipe), or
 - IV.B.3.b.(ii) Has a rated manufacturer's capacity of 7,571 liters (2,000 gallons) or more and was installed before November 7, 1973.
- IV.B.3.c. Tanks equipped with a submerged fill pipe shall meet the specifications of Appendix B.
- IV.B.3.d. The owner or operator of storage tanks at a gasoline dispensing facility must install and operate one or more of the following
 - IV.B.3.d.(i) A vapor collection system designed and operated in accordance with a vapor-tight line from the storage tank to delivery vessel.
 - IV.B.3.d.(ii) A refrigerator-condensation system or equivalent designed to recover at least 90 percent by weight of the organic compounds in the displaced vapor.
- IV.B.3.e. The owner or operator shall ensure that operating procedures are used so that gasoline cannot be transferred into the tank unless the vapor collection system is installed and operated to ensure the system is leak-tight during gasoline transfer.
- IV.B.3.f. The vapor collection system shall meet the specifications of Appendix B and applicable requirements of Section VII.

IV.B.3.g. Control devices shall meet the applicable requirements, including recordkeeping, of Regulation Number 25, Part B, Sections I.A.3.a., b., c., and e., and I.A.8.a. and b.

IV.B.3.h. The applicable EPA reference methods 1 through 4, and 25, of 40 CFR Part 60 (March 23, 2021) shall be used to determine the efficiency of control devices.

IV.C. Transfer of Petroleum Liquid

IV.C.1. Exemptions

Transfer operations involving petroleum liquid with true vapor pressures at 20°C of less than 78 torr (1.5 psia) or greater than 570 torr (11.0 psia) shall be exempt from the provisions of this Section IV.C.

IV.C.2. Loading Facilities Classified as Terminals

IV.C.2.a. A terminal is defined as a petroleum liquid storage and distribution facility that has an average daily throughput of more than 76,000 liters of gasoline (20,000 gallons), which is loaded directly into transport vehicles. A rolling, 30-day average of throughput shall be used to determine the applicability of this Section IV.C.2.

IV.C.2.b. The owner or operator of a terminal subject to this section shall equip the terminal with proper loading equipment and shall follow the loading procedures listed:

IV.C.2.b.(i) Install dry-break loading couplings to prevent petroleum liquid loss during uncoupling from vehicles.

IV.C.2.b.(ii) Install a vapor collection and disposal system which gathers vapor transferred from vehicles being loaded. The system shall include devices to prevent the release of vapor from vapor recovery hoses not in use.

IV.C.2.b.(iii) Use operating procedures to ensure that petroleum liquid cannot be transferred unless the vapor collection equipment is in use.

IV.C.2.b.(iv) Provide for the prevention of overfilling of transport vehicles with loading pump shut-offs, set stop meters, or comparable equipment.

IV.C.2.b.(v) Operate all recovery and disposal equipment at a back pressure less than the pressure relief valve setting of transport vehicles.

IV.C.2.b.(vi) Prevent the release of petroleum liquid on the ground from transport vehicles. Provision shall be made to remove any undelivered petroleum liquid with closed drainage devices.

IV.C.2.b.(vii) Maintain and operate final recovery and disposal equipment or control devices so as to emit no more than 80 milligrams of volatile organic compounds per liter of gasoline being loaded. Such disposal devices shall be approved by the Division.

IV.C.2.b.(viii) Prevent loading of petroleum liquid into transport vehicles which do not have valid leak-tight test certification as required in Section IV.D.

IV.C.2.b.(ix) Follow all control procedures to prevent leaks as specified in Section VII.

IV.C.2.c. Control devices shall meet the applicable requirements, including recordkeeping of Regulation Number 25, Part B, Sections I.A.3.a., b., c., and e., and I.A.8.a. and b.

IV.C.2.d. The applicable methods of 40 CFR 60.503 (December 19, 2003), or EPA reference methods 1 through 4, 25A, and 25B of 40 CFR Part 60 (March 23, 2021) shall be used to determine the efficiency of control devices.

IV.C.2.e. The method set forth in Appendix A of "Control of Hydrocarbons from Tank Truck Gasoline Loading Terminals" October 1977, EPA-450/2-77-026 shall be used to test emission points other than control devices.

IV.C.3. Loading Facilities Classified as Bulk Plants

IV.C.3.a. A bulk plant is defined as a petroleum liquid storage and distribution facility that has an average daily throughput of 76,000 liters of gasoline (20,000 gallons) or less, which is loaded directly into transport trucks. (As used herein, "bulk plant" does not include service stations nor separate operations within petroleum liquid distribution facilities which pump only into fuel tanks fueling motor vehicles. Both such operations are regulated by Section IV.B.3.). A rolling 30-day average of throughput shall be used to determine the applicability of this regulation.

IV.C.3.b. The owner or operator of a bulk storage plant subject to this section shall install a vapor balance system to return vapors to the incoming transport trucks during the filling of tanks controlled under

Section IV.B.3. The vapor balance system must be designed and operated in accord with the provisions of Appendix C.)

IV.C.3.c. The owner or operator of a bulk plant which serves storage tanks which are required to collect and recover vapor as prescribed in Section IV.B.3. shall:

IV.C.3.c.(i) Install and operate vapor collection and return equipment on any transport vehicles used to deliver to controlled tanks, and

IV.C.3.c.(ii) Install and operate vapor collection and return equipment at loading facilities to collect vapors during loading of tank compartments of outbound transport trucks and return these vapors to the bulk plant storage tanks, using a vapor balance system.

IV.C.3.c.(iii) Assure that transport trucks and loading facilities conform to the applicable provisions of Sections IV.C.2. and IV.C.4.

IV.C.3.d. The owner or operator of a bulk plant which serves only storage tanks exempted from the provisions of Section IV.B.3.b. by reason of their small size or location in an attainment area shall load outbound transport trucks using equipment that provides for top loading of the petroleum liquid into the vehicle tank compartments through an extended fill tube which reaches within 15.24 cm (6 in.) of the bottom of the tank compartment.

IV.C.3.e. The owner or operator of a bulk plant subject to this section shall ensure that petroleum liquid cannot be transferred unless the vapor collection equipment is in use.

IV.C.3.f. The owner or operator of a bulk plant subject to this section shall follow all procedures to prevent leaks as specified in Section VII.

IV.C.4. Transport Vehicles

IV.C.4.a. Rail cars shall be loaded only at facilities which allow for the following:

IV.C.4.a.(i) A submerged fill pipe which reaches within 15.24 cm (6 in.) of the bottom of the tank.

IV.C.4.a.(ii) Vapor collection and/or disposal equipment designated and operated to recover vapors displaced during the loading of the rail car.

IV.C.4.a.(iii) A vapor-tight seal around the tank car hatch and the loading equipment.

IV.C.4.b. The owner or operator of petroleum transport trucks which serve locations required to be equipped with vapor recovery equipment shall load only at facilities capable of disposing of collected vapors. The owner or operator shall assure that such vehicles possess the proper equipment and that work practices are followed so that:

IV.C.4.b.(i) Dry-break loading and unloading nozzles are used and are compatible with those required at loading facilities.

IV.C.4.b.(ii) Vapor recovery hoses are connected at all times during unloading or loading of petroleum distillate.

IV.C.4.b.(iii) Transport trailers and vehicle tanks are operated and maintained to prevent detectable hydrocarbon vapor loss during loading, transport and delivery.

IV.C.4.b.(iv) Compartment dome lids are closed and locked during transfers of petroleum liquid. Such lids may be opened for the purpose of certifying the accuracy of a delivery only prior to and after such delivery.

IV.C.4.b.(v) Hoses, couplings, and valves are maintained to prevent dripping, leaking, or other liquid or vapor loss during loading or unloading.

IV.D. Control of Volatile Organic Compound Leaks from Gasoline Transport Trucks

IV.D.1. General Provisions

IV.D.1.a. Applicability

This section is applicable to all gasoline transport trucks equipped for gasoline vapor collection which receive or dispense gasoline at terminals, bulk plants, or gasoline dispensing facilities located in the nonattainment areas.

IV.D.1.b. Definitions

For the purpose of this section, the following definitions apply:

IV.D.1.b.(i) "Gasoline Transport Truck" means a tank truck or tank trailer equipped with a storage tank and used for the transport of gasoline from sources of supply to stationary

storage tanks of gasoline dispensing facilities (e.g., service stations), bulk gasoline plants, or gasoline terminals.

IV.D.1.b.(ii) "Vapor Collection System" means a vapor transport system which uses direct displacement by the gasoline being transferred to force vapors from the vessel being loaded into a vessel being unloaded or into a vapor holding tank.

IV.D.2. Provisions for Specific Processes

IV.D.2.a. No terminal operator, when monitoring the gasoline loading operation and no owner or operator of a gasoline transport truck shall allow a gasoline transport truck subject to this Section IV.D. to be filled with a VOC with Reid Vapor Pressure of 4.0 or greater unless the gasoline tank truck:

IV.D.2.a.(i) Is tested annually according to the test procedure in EPA Method 27 (40 CFR Part 60, Appendix A-8) (October 17, 2000). Except as provided in Section IV.D.2.c., testing must be completed prior to the onset of the summer ozone season (test October through April). In addition, a visual inspection, as detailed in Section IV.D.3.b. must be performed at least once every six months.

IV.D.2.a.(i)(A) The test must be conducted using a time period (t) for the pressure and vacuum tests of 5 minutes.

IV.D.2.a.(i)(B) The initial pressure (P_i) for the pressure test must be 460 mm H₂O (18 in. H₂O), gauge.

IV.D.2.a.(i)(C) The initial vacuum (V_i) for the vacuum test must be 150 mm H₂O (6 in. H₂O), gauge.

IV.D.2.a.(i)(D) The maximum allowable pressure and vacuum changes must not exceed the values in Table 1.

IV.D.2.a.(i)(E) After completing the tests under Sections IV.D.2.a.(i)(A) through (D), the tank's internal vapor valve must be pressure tested.

IV.D.2.a.(i)(E)(1) Use the procedures in EPA Method 27 (40 CFR Part 60, Appendix A-8) (October 17, 2000) to repressurize the tank to 460 mm H₂O (18 in. H₂O), gauge.

IV.D.2.a.(i)(E)(2) Close the tank's internal vapor valve(s), thereby isolating the vapor return line and manifold from the tank.

IV.D.2.a.(i)(E)(3) Relieve the pressure in the vapor return line to atmospheric pressure, then reseal the line.

IV.D.2.a.(i)(E)(4) After 5 minutes, record the gauge pressure in the vapor return line and manifold. The maximum allowable 5-minute pressure increase is 130 mm H₂O (5 in. H₂O).

Table 1 - Allowable Cargo Tank Test Pressure or Vacuum Change		
Cargo tank or compartment capacity, liters (gal)	Annual certification-allowable pressure or vacuum change in 5 minutes, mm H ₂ O (in. H ₂ O)	Allowable pressure change in 5 minutes at any time, mm H ₂ O (in. H ₂ O)
9,464 or more (2,500 or more)	25 (1.0)	64 (2.5)
9,463 to 5,678 (2,499 to 1,500)	38 (1.5)	76 (3.0)
5,679 to 3,785 (1,499 to 1,000)	51 (2.0)	89 (3.5)
3,782 or less (999 or less)	64 (2.5)	102 (4.0)

IV.D.2.a.(ii) Passes a retest within twenty (20) days if it does not meet the criteria of Section IV.D.2.a.(i).

IV.D.2.b. Monitoring

IV.D.2.b.(i) The Division may, at any time, monitor a gasoline tank truck vapor collection system to confirm continued compliance with Section IV.D.2.a.

IV.D.2.b.(ii) Within fifteen (15) days after an exceedance is detected a tank shall pass a pressure/vacuum test per EPA Method 27 (40 CFR Part 60, Appendix A-8 (October 17, 2000)).

IV.D.2.c. Alternative Testing Schedule

IV.D.2.c.(i) Beginning February 14, 2023, the testing required in Section IV.D.2.a. may be completed during the months May through September if the testing facility controls the purged vapors from all gasoline transport trucks tested or purged for other reasons (e.g., maintenance, repair) at the facility by routing emissions to and operating air pollution control equipment that has a VOC control efficiency of at least 95%.

IV.D.2.c.(i)(A) Emission controls must be utilized on all vessels purged throughout the entire calendar year.

IV.D.2.c.(i)(B) If a combustion device is used, it must have a design destruction efficiency of at least 98%.

IV.D.2.c.(i)(C) All combustion devices used to control emissions must be equipped with and operate an auto-igniter.

IV.D.2.c.(i)(D) Weekly visual inspections

IV.D.2.c.(i)(D)(1) Inspection or monitoring of each combustion device to ensure that it is operating, including that the pilot light is lit and the auto-igniter is properly functioning.

IV.D.2.c.(i)(D)(2) Inspection or monitoring of each combustion device to ensure that the valves for the piping of gas to the pilot light are open and functioning properly.

IV.D.2.c.(i)(D)(3) Inspection or monitoring of each combustion device to ensure the burner tray is not visibly clogged.

IV.D.2.c.(i)(D)(4) Inspection of each combustion device for the presence or absence of smoke. If smoke is observed, either the equipment must be immediately shut-in to investigate the potential cause for smoke and perform repairs, as necessary, or EPA Method 22 must be conducted to determine whether visible emissions are present for a period of at least one (1) minute in fifteen (15) minutes.

IV.D.2.c.(i)(E) Enclosed combustion devices must be performance tested every five (5) years, beginning five (5) years after the date the combustion device

commenced operation or the date of the previous performance test.

IV.D.2.c.(ii) Recordkeeping. The testing facility must maintain records for a period of five (5) years and make them available to the Division upon request.

IV.D.2.c.(ii)(A) Records of gasoline transport truck vapor purging events including the date and duration of the purging event and the reason for the vapor purging (e.g., testing, maintenance, repair).

IV.D.2.c.(ii)(B) Records of the inspections required in Section IV.D.2.c.(i)(C) including the time and date of each inspection, a description of any problems observed, a description and date of any corrective action(s) taken, and name of employee or third party performing corrective action(s).

IV.D.2.c.(ii)(C) Records of the date and result of any EPA Method 22 test or investigation.

IV.D.2.c.(ii)(D) Records of performance tests including the date of the test, a copy of the test protocol followed, documentation of the methods and results of the test, and the date and description of any actions taken in response to a failed test.

IV.D.3. Testing and Monitoring

IV.D.3.a. The owner or operator of a gasoline transport truck shall at their own expense, demonstrate compliance with Section IV.D.2, by methods of EPA Method 27 (40 CFR Part 60, Appendix A-8) (October 17, 2000). All tests shall be made by, or under the direction of, a person qualified by training and/or experience in the field of air pollution testing or gasoline transport truck maintenance.

IV.D.3.b. The owner or operator of a gasoline transport truck must conduct a visual inspection of the gasoline transport truck at least once every six months. The entire tank, including domes, dome vents, cargo tank, piping, hose connections, hoses and delivery elbows, must be inspected for wear, damage, or misadjustment that could be a potential leak source. Inspect all rubber fittings except those in piping which are not accessible. Any part found to be defective must be adjusted, repaired, or replaced as necessary.

IV.D.4. Recordkeeping and Reporting

IV.D.4.a. The owner or operator of a gasoline transport truck subject to this Section IV.D. shall maintain records of all certification testing and repairs. The records shall identify the gasoline transport truck, the date of the test or repairs and, if applicable, the type of repair and the date of retest. The written record shall include entries of any pre-test repairs, adjustments, or modifications. These shall also include the part name, number, and vendor name of any part removed and of any part installed. The records shall be maintained in legible, readily available form for at least two (2) years after the date the testing or repair was completed and shall be made available to the Division for inspection upon request.

IV.D.4.b. The records of certification tests required by Section IV.D.2.a. must, as a minimum, contain all of the following entries:

IV.D.4.b.(i) The gasoline transport truck owner's name and address;

IV.D.4.b.(ii) The gasoline transport truck/tank identification number;

IV.D.4.b.(iii) The nature of repair work and when performed in relation to vapor tightness testing.

IV.D.4.b.(iv) The following data for each test:

IV.D.4.b.(iv)(A) Test pressure.

IV.D.4.b.(iv)(B) Pressure or vacuum change, mm of water.

IV.D.4.b.(iv)(C) Time period of test.

IV.D.4.b.(iv)(D) Number of leaks found with instrument.

IV.D.4.b.(iv)(E) Leak definition.

IV.D.4.b.(v) The size of each of the compartments within the tank and whether such compartment was manifolded or was tested separately during pressure and vacuum tests.

IV.D.4.b.(vi) At the top of each report page shall be the company name and the date and location of the test results recorded on that page.

IV.D.4.b.(vii) Name and title of the person conducting the test.

IV.D.4.b.(viii) For the vapor valve test required in Section IV.D.2.a.(ii)(A), the initial test pressure and time of reading.

IV.D.4.c. The records of the visual inspections required by Section IV.D.3.b.

IV.D.4.d. The owner or operator of a gasoline transport truck subject to this regulation must annually certify to the Division that the gasoline transport truck has been tested by the applicable method(s) referenced in Section IV.D.3. The certification must include:

IV.D.4.d.(i) The name and address of the company and the name and telephone number of responsible company representative over whose signature the certification is submitted; and,

IV.D.4.d.(ii) A copy of the information recorded to comply with Section IV.D.4.b.

IV.D.4.e. The records of certification tests must be kept with the tank or at the transport company office at all times and must be shown to Division personnel upon their request. Copies of all records and reports required by the provisions of this Section IV.D. must be made available to the Division upon oral or written request.

V. Crude Oil

V.A. General Exemptions

V.A.1. Storage tanks of 151,412 liters (40,000 gallons) or less used to store crude oil is exempt from the provisions of this section.

V.A.2. Storage tanks with capacities of less than 1,590 cubic meters (10,000 barrels) used to store crude oil and condensate prior to lease custody transfer are exempt from the provisions of Regulation Number 24.

V.B. Equipment

Pumps and compressors handling crude oil shall be subject to the provisions of Section IV.A.

V.C. Storage

Except as provided in Section V.A.2., crude oil stored in tanks greater than 151,412 liters (40,000 gallons) shall be subject to the provisions of Sections IV.B.1.b. and IV.B.2.

VI. Petroleum Processing and Refining

VI.A. Wastewater (Oil/Water) Separators

VI.A.1. Definitions

VI.A.1.a. "Forebays" mean the primary sections of a wastewater separator.

VI.A.1.b. "Wastewater (oil/water) separator" means any device or piece of equipment which utilizes the difference in density between oil and water to remove oil and associated chemicals from water, or any device, such as a flocculation tank, clarifier, etc., which removes petroleum derived compounds from wastewater.

VI.A.2. The owner or operator of any wastewater (oil/water) separators at a petroleum refinery shall:

VI.A.2.a. Equip the forebays and separator sections of the wastewater separators with one or more of the following emission control devices, ensuring that such device is properly installed, in good working order and properly maintained:

VI.A.2.a.(i) A solid cover with all openings sealed and the liquid contents totally enclosed.

VI.A.2.a.(ii) A pontoon-type or double-deck type floating roof, or internal floating cover. The floating roof or cover must rest on the surface of the liquid contents and be equipped with a closure seal or seals to close the space between the edge of the floating roof (or cover) and the wall(s) of the compartment.

VI.A.2.a.(iii) A vapor recovery system consisting of a vapor gathering device capable of collecting the volatile organic compound vapors discharged and a vapor disposal device capable of processing such volatile organic vapors so as to prevent their emission into the atmosphere.

VI.A.2.a.(iii)(A) Control devices shall meet the applicable requirements, including recordkeeping, of Regulation Number 25, Part B, Sections I.A.3.a., b., c., and e., and I.A.8.a. and b.

VI.A.2.a.(iii)(B) The applicable EPA reference methods 1 through 4, and 25, of 40 CFR Part 60 (March 23, 2021) shall be used to determine the efficiency of control devices.

- VI.A.2.b. Equip all openings in covers, separators, and forebays with lids or seals such that the lids or seals are in the closed position at all times except when in actual use. Access for gauging and sampling shall be minimized.

VI.B. Emissions from Petroleum Refineries

VI.B.1. Definitions

- VI.B.1.a. "Firebox" means the chamber or compartment of a boiler or furnace in which materials are burned but does not mean the combustion chamber of an incinerator.
- VI.B.1.b. "Turnaround" means the procedure of shutting a refinery unit down after a run to do necessary maintenance and repair work and then putting the unit back on stream.

VI.B.2. Process unit turnarounds

The owner or operator of a petroleum refinery shall develop and submit to the Division for approval a detailed procedure for minimization of volatile organic compound emissions during process unit turnaround. As a minimum, the procedure shall provide for:

- VI.B.2.a. Depressurization venting of the process unit or vessel to a vapor recovery system, or to a flare or firebox which assures at least 90% combustion efficiency;
- VI.B.2.b. No emission of volatile organic compounds from a process unit or vessel until its internal pressure is 17.2 psia or less; and
- VI.B.2.c. Recordkeeping of the following items. Records shall be kept for at least two years and shall be made available to the Division for review upon request.
 - VI.B.2.c.(i) Every date that each process unit is shut down,
 - VI.B.2.c.(ii) The approximate vessel volatile organic compound concentration when the volatile organic compounds were first discharged to the atmosphere, and
 - VI.B.2.c.(iii) The approximate total quantity of volatile organic compounds emitted to the atmosphere.

VI.B.3. Venting of blowdown systems and safety pressure relief valves

All blowdown systems, process equipment vents, and pressure relief valves shall be vented to a vapor recovery system, or to a flare or firebox which assures at least 90% combustion efficiency.

VI.B.4. Vacuum-Producing Systems

VI.B.4.a. The owner or operator of any vacuum-producing system at a petroleum refinery shall not permit the emission of any noncondensable volatile organic compounds from the condensers, hot wells or accumulators of the system. This emission limit shall be achieved by:

VI.B.4.a.(i) Venting the noncondensable vapors to a flare or other combustion device, or,

VI.B.4.a.(ii) Compressing the vapors and adding them to the refinery fuel gas.

VI.B.5. All sampling, testing, and measuring ports, hatches, and access openings shall be kept in a closed sealed position except during actual sampling or access.

VI.B.6. Control devices shall meet the applicable requirements, including recordkeeping, of Regulation Number 25, Part B, Sections I.A.3.a., b., c., and e., and I.A.8.a. and b.

VI.B.7. The applicable EPA reference methods 1 through 4, and 25, of 40 CFR Part 60 (March 23, 2021), shall be used to determine the efficiency of control devices.

VI.C. Petroleum Refinery Equipment Leaks

VI.C.1. Definitions

For the purpose of this section, the following definitions apply:

VI.C.1.a. "Accessible Component" means a component which can be reached, if necessary, by safe and proper use of portable ladders such as are acceptable to OSHA, as well as by built-in ladders and walkways. "Accessible" also includes components which can be reached by the safe use of an extension on the monitoring probe.

VI.C.1.b. "Component" means any piece of equipment, which has the potential to leak volatile organic compounds when tested in the manner described in Section VI.C.3. These sources include, but are not limited to, pumping seals, compressor seals, seal oil degassing vents, pipeline valves, flanges and other connections, pressure

relief devices, process drains, and open ended pipes. Excluded from these sources are valves which are not externally regulated.

VI.C.1.c. "Gaseous Service" means equipment which processes, transfers or contains a volatile organic compound or mixture of volatile organic compounds in the gaseous phase.

VI.C.1.d. "In Heavy VOC Liquid Service" means that the piece of equipment is not in gaseous service or in light VOC liquid service.

VI.C.1.e. "In Light Liquid VOC Service" Equipment is in light liquid service if the following conditions apply:

VI.C.1.e.(i) the true vapor pressure of one or more of the components is greater than 0.3 kPa at 20°C. True vapor pressures may be obtained from standard reference texts or may be determined by ASTM D-2879.

VI.C.1.e.(ii) the total concentration of the pure components have a true vapor pressure greater than 0.3 kPa at 20°C, is equal to or greater than 20 percent by weight; and

VI.C.1.e.(iii) the fluid is a liquid at operating conditions.

VI.C.1.f. "Refinery Unit" means a set of components which are a part of a basic process operation, such as, distillation, hydrotreating, cracking, or reforming of hydrocarbons.

VI.C.1.g. "Water Draw" means a routinely used valve or system employing a valve which allows non-VOC material (usually water) to be separated from VOC.

VI.C.2. Provisions for Specific Processes

VI.C.2.a. The owner or operator of a petroleum refinery complex subject to this regulation must:

VI.C.2.a.(i) Develop a monitoring program consistent with the provisions in Section VI.C.3.

VI.C.2.a.(ii) Conduct a monitoring program consistent with the provisions in Section VI.C.4.a.

VI.C.2.a.(iii) Record all leaking components which have a VOC concentration exceeding 10,000 ppmv when tested according to Section VI.C.3., and place an identifying tag on each component consistent with the provisions in Section VI.C.4.a.(iii).

unless the pipe or line is sealed with a second valve, a blind flange, a plug, or a cap. The sealing device may be removed only when a sample is being taken or when the valve is otherwise in use.

VI.C.2.c. The Division, at its discretion, may require early unit turnaround based on the number and severity of tagged leaks awaiting turnaround provided:

VI.C.2.c.(i) The requirement does not exceed reasonable available control technology due to cost per ton of emissions reduction achieved by the early turnaround or other reasonable analysis.

VI.C.2.c.(ii) The Division provides the owner or operator of a petroleum refinery with written notification at least 180 days before requiring an early turnaround. The owner or operator will have 30 days from the date of the Division's notification to contest the requirement by submitting a demonstration that the requirement is beyond reasonable available control technology. If no demonstration is made, it will be assumed the requirement is reasonable. If a demonstration is submitted by the owner or operator, the Division will either approve the demonstration or disapprove the demonstration with a justification regarding the disapproval within 30 days of the date the demonstration is submitted to the Division.

VI.C.2.c.(iii) The requirement is not contested by the owner or operator. Should the requirement be contested, the requirement for early unit turnaround will be delayed until 180 days after the demonstration discussed in Section VI.C.2.c.(ii) is disapproved by the Division.

VI.C.2.d. Piping valves and pressure relief valves in gaseous VOC service shall be marked in some manner that will be readily obvious to both refinery personnel performing monitoring and the Division, to identify them as components which are monitored quarterly.⁷

VI.C.3. Testing and Monitoring Procedures

VI.C.3.a. Testing and calibration procedures to determine compliance with this regulation must be consistent with EPA reference method 21 of 40 CFR Part 60 (October 7, 2000). The reference compound may be methane or hexane. A leak is defined as a reading of 10,000 ppmv of the reference compound.

VI.C.3.b. (State Only) Beginning June 30, 2027, a leak identified at a non-pump component is defined as a reading of 1,000 ppmv or greater of the reference compound and beginning June 30, 2028, a

leak identified at a non-pump component is defined as a reading of 500 ppmv or greater of the reference compound.

VI.C.3.c. (State Only) Beginning June 14, 2026, a leak identified at a pump is defined as a reading of 2,000 ppmv or greater of the reference compound.

VI.C.4. Monitoring, Recordkeeping, Reporting

VI.C.4.a. Monitoring

VI.C.4.a.(i) The owner or operator of a petroleum refinery subject to this regulation shall conduct a monitoring program consistent with the following provisions:

VI.C.4.a.(i)(A) Monitor yearly by the method referenced in Section VI.C.3., all:

VI.C.4.a.(i)(A)(1) Pump seals; and

VI.C.4.a.(i)(A)(2) Piping valves in light liquid VOC service; and

VI.C.4.a.(i)(A)(3) Process drains; and

VI.C.4.a.(i)(A)(4) Heat-exchanger body flanges; and

VI.C.4.a.(i)(A)(5) Other accessible flanges in VOC service.

VI.C.4.a.(i)(A)(6) Components in heavy liquid VOC service are exempt from requirements of this Section VI.C.4.a.(i)(A).

VI.C.4.a.(i)(B) Monitor quarterly by the method referenced in Section VI.C.3., all:

VI.C.4.a.(i)(B)(1) Compressor seals; and

VI.C.4.a.(i)(B)(2) Piping valves in gaseous service; and

VI.C.4.a.(i)(B)(3) Pressure relief valves in gaseous service.

VI.C.4.a.(i)(C) Monitor at least weekly by visual methods all pump seals.

VI.C.4.a.(i)(D) Monitor within 24 hours with a VOC detector and make record of any component from which VOC liquids are observed leaking.

VI.C.4.a.(i)(E) Components in heavy liquid VOC service shall be monitored by the method referenced in Section VI.C.3. within five days if evidence of a potential leak is found by visual, audible, olfactory, or any other detectable method.

VI.C.4.a.(ii) Inaccessible valves and flanges must be monitored annually or, as a minimum, at unit shutdown using the procedures of VI.C.2.a.(viii). Pressure relief devices which are connected to an operating flare header or vapor recovery device, storage tank valves, and valves that are not externally regulated are exempt from the monitoring requirements in Section VI.C.4.a.(i).

VI.C.4.a.(iii) The owner or operator of a petroleum refinery, upon the detection of a leaking component as defined in Section VI.C.2.a.(iii) (State Only) and Sections VI.C.2.a.(iv) through VI.C.2.a.(vi), must affix a weatherproof and readily visible tag, bearing an identification number and the date the leak is located, to the leaking component. This tag must remain in place until the leaking component is repaired. In addition, the owner or operator must log the leak (including those leaks immediately repaired), per the requirements of Sections VI.C.4.b.(i) through (iii).

VI.C.4.b. Recordkeeping

VI.C.4.b.(i) The owner or operator of a petroleum refinery must maintain a leaking components monitoring log which must contain at a minimum, the following data:

VI.C.4.b.(i)(A) The name of the process unit where the component is located.

VI.C.4.b.(i)(B) The type of component (e.g., valve, seal).

VI.C.4.b.(i)(C) The tag number of the component.

VI.C.4.b.(i)(D) The date on which a leaking component is discovered.

VI.C.4.b.(i)(E) The date on which a leaking component is repaired.

VI.C.4.b.(i)(F) The date and instrument reading found during the recheck procedure subsequent to repairing a leaking component.

VI.C.4.b.(i)(G) A record of the calibration of the monitoring instrument.

VI.C.4.b.(i)(H) Those leaks that cannot be repaired until turnaround.

VI.C.4.b.(i)(I) The total number of components checked and the total number of components found leaking.

VI.C.4.b.(i)(J) The total number of components subject to Section VI.C.2.a.(viii) which upon retest were still leaking as defined in Section VI.C.3.

VI.C.4.b.(ii) Copies of the monitoring log shall be retained by the owner or operator for a minimum of two (2) years after the date on which the record was made or report prepared.

VI.C.4.b.(iii) Copies of the monitoring log shall be made available to the Division upon oral or written request.

VI.C.4.c. Reporting

The owner or operator of a petroleum refinery, upon the completion of each yearly and/or quarterly monitoring procedure, shall:

VI.C.4.c.(i) Submit a report to the Division by the 15th day of February, May, August, and November that lists all leaking components that were located during the previous three (3) calendar months (quarter), but not repaired within fifteen (15) working days, all leaking components awaiting unit turnaround, the total number of components inspected, and the total number of components found leaking.

VI.C.4.c.(ii) Submit a signed statement with the report attesting to the fact that, with the exception to those leaking components listed in Section VI.C.4.b.(i)(H), all monitoring and repairs were performed as stipulated in the monitoring program.

VII. Control of Volatile Organic Compound Leaks from Vapor Collection Systems and Vapor Control Systems Located at Gasoline Terminals, Gasoline Bulk Plants, and Gasoline Dispensing Facilities

VII.A. General Provisions

VII.A.1. Applicability

This section is applicable to all gasoline terminals, gasoline bulk plants, and gasoline dispensing facilities (e.g., service stations) which are located in ozone nonattainment areas and which must have a vapor collection system pursuant to Section IV. and other applicable rules.

VII.A.2. Exemptions

This section is not applicable to those operations involving transfer of gasoline from gasoline dispensing facilities to motor vehicle fuel tanks nor to other dispensing operations at such facilities.

VII.A.3. Definitions

For the purpose of this section, the following definitions apply:

VII.A.3.a. "Gasoline Dispensing Facility" means any site where gasoline is dispensed to motor vehicle fuel tanks from stationary storage tanks, (e.g., service stations, fleet pumps, etc.)

VII.A.3.b. "Gasoline Transport Truck" means tank trucks or trailers equipped with a storage tank and used for the transport of gasoline from sources of supply to stationary storage tanks of gasoline dispensing facilities (e.g., service stations), bulk gasoline plants or gasoline terminals.

VII.A.3.c. "Vapor Collection System" means a vapor transport system which uses direct displacement by the gasoline being transferred to force vapors from the vessel being loaded into either a vessel being unloaded or a vapor holding tank.

VII.B. Specific Provisions

VII.B.1. The operator of a vapor collection system at a facility subject to the provisions of this section shall operate the vapor collection system and the gasoline loading equipment in a manner that prevents:

VII.B.1.a. Gauge pressure from exceeding 33.6 torr (18 inches of H₂O) and vacuum from exceeding gauge pressure of minus 11.2 torr (minus 6 inches of H₂O) at the point where the vapor return line on the truck connects with the vapor collection line of the facility.

VII.B.1.b. A reading equal to or greater than 100 percent of the lower explosive limit (LEL, measured as propane) at 2.5 centimeters from a known or potential leak source when measured by the procedures described in Appendix B of "Control of Organic Compound Leaks from Gasoline Tank Trucks and Vapor Collection

Systems,” December 1978, EPA-450/2-78-051, during loading or unloading operations at gasoline dispensing facilities, bulk plants, and terminals.

VII.B.1.c. Avoidable liquid or vapor leaks from the system during loading or unloading operations at gasoline dispensing facilities, bulk plants, and terminals.

VII.B.1.d. Division representatives may monitor for excessive back pressure as defined by Section VII.B.1.a. and vapor leakage as is defined by Section VII.B.1.b. or by detection methods incorporating sight, sound, and smell.

VII.B.2. Repairs and Modifications

VII.B.2.a. The operator shall within fifteen (15) days, repair and retest a vapor collection system that exceeds the pressure limits (Section VII.B.1.a.), excepting that;

VII.B.2.b. Should an applicable facility require modification or repairs that will take longer than fifteen (15) days to complete, the operator shall submit to the Division for approval a schedule which includes dates of commencement and completion.

Appendix B Criteria for Control of Vapors from Gasoline Transfer to Storage Tanks

- I. Drop Tube Specifications. Submerged fill is specifically required. The drop tube must extend to within 15.24 cm (6 in.) of the tank bottom.
- II. Vapor Hose Return. Vapor return line and any manifold must be minimum 7.6 cm (3 in.) ID. All tanks must be provided with individual overfill protection. (Liquid must not be allowed in the vent line or vapor recovery line.) Disconnect on liquid line should assure that all liquid in the hose is drained into the storage tank. The requirements for overfill protection as specified may be waived for existing storage tanks when it is demonstrated to the satisfaction of the appropriate local Fire Marshal, and where applicable, the State Oil Inspection Office that the installation of overfill protection devices on existing tanks is physically not possible.
- III. Size of Vapor Line Connections. For separate vapor lines, nominal three inch (7.6 cm) or larger connections must be utilized at the storage tank and truck. However, short lengths of 2-inch (5.1 cm) vertical pipe no greater than 91.4 cm (3 ft.) long are permissible if the fuel delivery rate is less than 400 gallons per minute.

Where concentric (coaxial) connections are utilized, a 45 cm² (7 sq. in.) area for vapor return shall be provided. Four-inch concentric designs are acceptable only

when using a venturi-shaped outer tube or where normal drop rate of 1,700 liters per minute (450 gpm) is reduced by at least 25%. Six-inch (15.24 cm) risers should be installed in new stations with concentric connections.

- IV. Type of Liquid Fill Connection. Vapor tight caps are required for the liquid fill connection for all systems. A positive closure utilizing a gasket is necessary to prevent vapors from being emitted at ground level. Cam-lock closures meet this requirement. Dry break closures are preferred.
- V. Tank Truck Inspection. Tank trucks are specifically required to be vapor-tight and to have valid leak-tight certification. The visual inspection procedure must be conducted at least once every six months to ensure properly operating manifolding and relief valves, using the test procedure of Section IV.D.3.b.
- VI. Dry Break on Underground Tank Vapor Riser. Dry-break closures are required to assure transfer of displaced vapors to the truck and to prevent ground-level, gasoline-vapor emissions caused by failure to connect the vapor return line to the underground tanks (closure on riser to mate with opening on hose). These devices keep the tank sealed until the hose is connected to the underground tank. Concentric couplers without dry-breaks are required to have a dry-break on the vapor line connection to the coupler itself, rather than on the rise pipe from the storage tank. The liquid fill riser should be provided with a gap having a positive closure (threaded or latched).
- VII. Equipment Ensuring Vapor-Hose Connection during Gasoline Deliveries. An equipment system aboard the tank truck shall insure (barring deliberate tampering) that a vapor return hose is connected from the truck's vapor return line to the tank receiving gasoline.
- VIII. Vent Line Restriction Devices. Vent line restriction devices are required. If the liquid fill line were attached to the underground tank and the vapor return line were disconnected, then dry break closures would seal the vapor return path to the truck, forcing all vapors out the vent line. In such instances, a restriction device on this vent line greatly reduces fill rate, warning the operator that the vapor line is not connected.

Pressure/vacuum (PV) vent valves installed on the storage tank vent pipes. The pressure specifications for PV vent valves shall be: a positive pressure setting of 2.5 to 6.0 inches of water and a negative pressure setting of 6.0 to 10.0 inches of water.

- IX. Fire and Safety Regulations. All new or modified installations must comply in their entirety with all code requirements including NFPA, Pamphlet 30 (fiberglass is preferred for new manifold lines). For any questions concerning compliance, please contact State Oil Inspection or your local Fire Marshal.
- X. State Oil Inspection. Requirements of the State Oil Inspection office make accurate measurements of the liquid in the underground tank necessary. Vapor-

tight gauging devices will be required in all systems designed such that a pressure other than atmospheric will be held or maintained in the storage tank. The volume of liquid in the tanks maintained at atmospheric pressure may be determined with a stick through the submerged drop tube or through a separate submerged gauging tube extending to within 15.24 cm (6 in.) of the tank bottom.

Appendix C Criteria for Control of Vapors from Gasoline Transfer at Bulk Plants

I. Storage Tank Requirements:

- A. Drop Tube Specification: Underground tanks must contain a drop tube that extends to within six inches (15.24 cm) of the tank bottom. All top loaded above-ground tanks must contain a similar drop tube. Above-ground tanks using bottom loading, where the inlet is flush with the tank bottom, must meet the submerged fill requirement.
- B. Size of Vapor Lines from Storage Tanks to Loading Rack: See nomograph (Attachment 1). NOTE: Affected sources are free to choose a pipe diameter different from the one suggested by the nomograph if sufficient justification and documentation is presented.
- C. Pressure Relief Valves: All pressure relief valves and valve connections must be checked periodically for leaks, and be repaired as required. The relief valve pressures should be set in accordance with Sections 2-2.5.1 and 2-2.7.1 inclusive of the current National Fire Protection Agency Pamphlet Number 30.
- D. Liquid Level Check Port: Access for checking liquid level by other than a vapor-tight gauging system shall be vapor-tight when not being used. Tank level shall be checked prior to filling to avoid overfills.
- E. Miscellaneous Tank Openings: All other tank openings, e.g., tank inspection hatches, must be vapor tight when not being used, and must be closed at all times during transfer of fuel.
- F. Storage Tank Overfill Protection: Except for concentric (coaxial) delivery systems, underground tanks must have ball check valves (stainless steel ball). Tanks with concentric delivery systems must have Division-approved overfill protection, (e.g., cutoff pressure-switch in vent line).

II. Loading Rack Requirements:

- A. Loading Specification: A vapor-tight bottom-loading or top-loading system using submerged fill with a positive seal, e.g., the Wiggins (tm) system, is required. NOTE: Bulk plants delivering solely to exempt accounts are required to have submerged fill, but loading need not be vapor-tight.

- B. Dry-Break on Storage Tank Vapor Return Line: A dry-break is required to prevent ground-level gasoline vapor emissions during periods when gasoline transfer is not being made. This device keeps the tank sealed until the vapor return hose is connected.
- III. Tank Truck* Requirements:
- A. Vapor Return Modification: Tank trucks must be modified to recover vapors during loading and unloading operations. NOTE: Tank trucks making deliveries solely to exempt accounts do not require this modification. However, 97% submerged fill is required when top loading.
 - B. Loading Specifications: Bottom loading or top loading using submerged fill with a positive seal is required for tank trucks modified for vapor recovery. NOTE: When loading a tank truck with this modification without the vapor return hose connected (this is allowed at bulk plants servicing exempt accounts returning without collected vapors in the tank), the requirements of National Fire Protection Agency Pamphlet Number 385, "Loading and Unloading Venting Protection in Tank Vehicles, Section 2219, Paragraph c", must be met.
 - C. Vapor Return Hose Size: A minimum three-inch (7.6 cm) ID vapor return hose is required.
 - D. Tank Truck Inspection: Tank trucks are required to be vapor-tight and have valid leak-tight certification. Periodic visual inspection is necessary to insure properly operating manifolding and relief valves.

* The term "tank truck" is meant to include all trucks with tanks used for the transport of gasoline, such as tank wagons, account trucks and transport trucks.

PART C Statements of Basis, Specific Statutory Authority and Purpose

I. April 20, 2023

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

To improve the readability and usability of Regulation Number 7 and Regulation Number 22, the Commission adopted revisions restructuring and reorganizing the parts and sections.

Specific Statutory Authority

The Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq. (the State Air Act or the Act), specifically § 25-7-103.3, directs rule-making agencies, such as the Commission, to review their rules and consider whether the rule is necessary; whether the rule overlaps or duplicates other rules of the agency or with other federal, state, or local government rules; whether the rule is written in plain language and is easy to understand; whether the rule has achieved the desired intent and whether more or less regulation is necessary; whether the rule can be amended to give more flexibility, reduce regulatory burdens, or reduce unnecessary paperwork or steps while maintaining its benefits; whether the rule is implemented in an efficient and effective manner, including the requirements for the issuance of permits and licenses; whether a cost-benefit analysis was performed by the applicable rule-making agency; and whether the rule is adequate for the protection of the safety, health, and welfare of the state or its residents. Based on this review, the rule-making agency will determine whether the existing rules should be continued in their current form, amended, or repealed.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 24. The Commission reorganized Regulation Number 7 into four regulations: Part B became Regulation Number 24; Part C became Regulation Number 25; Part D remained in Regulation Number 7; and Part E became Regulation Number 26. The upstream oil and gas intensity and midstream combustion program provisions currently in Regulation Number 22 moved to Regulation Number 7. The manufacturing sector greenhouse gas provisions in Regulation Number 22 became a new Regulation Number 27.

To assist in tracking the history of the regulatory revisions, associated statements of basis and purpose, and restructured location, the Commission provides the following tracking table.

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
1995	Dec. 21	Clarify substances that are negligibly reactive VOCs.	Section II.B.	Part A, Section II.B.	Regulations 7 and 24-26, Part A
1996	Mar. 21	Revisions related to the maintenance demonstration.	Sections I.A.1. through I.A.4.; II.D.; II.E.	Part A, Sections I.A.1. through I.A.4.; II.D.; II.E.	Regulations 7 and 24-26, Part A

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
1996	Nov. 21	Updated NRVOC list. Removed control of VOC emissions from dry cleaning facilities using perchloroethylene.	Section XII.	NA	NA
1998	Oct. 15	Revisions specific to Gates Rubber Company.	Section II.F.	NA	Regulation 24-25, Part A
2001	Jan. 11	Correct discrepancies in posted versus adopted provisions.	Sections III.C.; IX.L.2.c.(1); X.D.2. through XI.A.3.	Part B, Section I.; Part C, Section I.; Part C, Sections II. through III.	Regulation 24, Part B; Regulation 25, Part B (fkna Part C)
2003	Nov. 20	Repealed provisions establishing a procedure for granting exemptions for de minimis sources and for approving alternative compliance plans.	Sections I.A.2. through I.A.4.; II.D.; II.E.	Part A, Sections I.A.1. through I.A.4.; II.D.; II.E.	Regulations 7 and 24-26, Part A
2004	Mar. 12	Revisions adopted in conjunction with the early action compact ozone action plan – control of emissions from condensate operation at oil and gas facilities, emissions from internal combustion engines, emissions from gas processing plants, and emissions from oil and gas operations dehydrators.	Sections I.A.; I.B.; XII.; XVI.	Part A, Section I.A.; Part A, Section I.B.; Part D, Section I.; Part E, Section I.	Regulations 7 and 24-26, Part A

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2004	Dec. 16	Revisions adopted in response to EPA comments (re practical enforceability) on the ozone action plan adopted 3/2004.	Sections I.A.; II.A.; XII.; XVI.;	Part A, Section I.A.; Part A, Section II.A.; Part D, Section I.; Part E, Section I.	Regulations 7 and 24-26, Part A
2006	Dec. 17	Expanding oil and gas condensate tank emission controls.	Section XII.	Part D, Section I.	Regulation 7, Part B (fkna Part D)
2006	Dec. 17	Reduce emissions from oil and gas operations and natural gas fired engines.	Sections I.A.1.b.; XVII.	Part A, Section I.A.; Part D., Section II. & Part E. Section I. (for engines)	Regulation 7, Part A and Part B (fkna Part D); Regulation 26, Part A and Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2008	Dec. 12	Expand VOC RACT requirements for 100 tpy sources and clarify how RACT requirements in Regulation Numbers 3 and 7 interact in the ozone nonattainment area. Make typographical and formatting changes. Revise oil and gas condensate tank and pneumatic controller requirements.	Title; Sections I.; II.; VI. through XIII.; XVII.; XVIII.; and Appendices A through F	Part A, Section I.; Part A, Section II.; Part B, Sections IV. through VI. & Part C, Sections I. through IV. & Part D, Section I.; Part D, Section II. and Part E, Section I. (for engines); Part D, Section III.; Part A, Appendix A. & Part B, Appendices B and C & Part C, Appendices D and E (formerly Appendix F)	Regulation 24, Part B; Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2011	Jan. 7	Include engine requirements in the Regional Haze SIP.	Outline; Sections I.; XVII.	Part A, Section I.; Part E, Section I.	Regulation 26, Part B (fkna Part E)
2012	Dec. 20	Address EPA comments on the June 2009 submittal. Revise state-only requirements for consistency.	Sections II.; XII.; XVII.	Part A, Section II.; Part D, Section I.; Part D, Section II.	Regulation 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2014	Feb. 23	Adopt additional oil and gas emission reduction requirements – auto-igniters, expand condensate tank controls, limit storage tank venting, expand dehydrator control, establish leak detection and repair program, limit venting during well maintenance and liquids unloading, expand pneumatic controller requirements.	Sections II.; XVII.; XVIII.	Part A, Section II.; Part D, Section II.; Part D, Section III.	Regulation 7, Part B (fkna Part D)
2016	Nov. 17	Adopt RACT requirements for industrial cleaning solvents, lithographic and letterpress printing, and specific major sources. Including existing combustion device auto-igniter and storage tank inspection requirements in the SIP. Adopting major source combustion equipment combustion process adjustment requirements and incorporate by reference NSPS and NESHAP for specific major sources.	Sections I.; X.; XII.; XIII.; XVI.; XIX.	Part A, Section I.; Part C, Section II.; Part D, Section I.; Part C, Section IV.; Part C, Section V.; Part E, Section III.	Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2017	Nov. 16	Adopt provisions based on recommendations in EPA's Oil and Gas Control Techniques Guideline. Revise state-only requirements for consistency.	Sections II.; XII.; XVII.; XVIII.	Part A, Section II.; Part D, Section I.; Part D, Section II.; Part D, Section III.	Regulation 7, Part B (fkna Part D)
2018	July 19	Adopt requirements for existing major source boilers, turbines, lightweight aggregate kilns, glass melting furnaces, engines.	Sections XVI.; XIX.	Part E, Section II., Part E, Section III.	Regulation 26, Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2018	Nov. 15	Adopt requirements for major source breweries and wood furniture manufacturing. Address EPA concerns with requirements for industrial cleaning solvents, metal furniture surface coating, and miscellaneous metal surface coating. Updated incorporation by reference dates.	Sections I.; II.; VI.; VIII.; IX.; X.; XII.; XIII.; XVI.; XVII.; XIX.; XX.; XXI.	Part A, Section I.; Part A, Section II.; Part B, Section IV.; Part B, Section VI.; Part C, Section I.; Part C, Section X.; Part D, Section I.; Part C, Section IV.; Part C, Section V.; Part D, Section II.; Part E, Section III.; Part E, Section IV. Part F	Regulation 24, Part B; Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E); Regulation 23 (fkna Part F)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2019	Dec. 19	Reorganized into Parts A through F. Replaced the SIP system-wide condensate tank control program with a fixed threshold storage tank control program. Increased state-only, state-wide storage tank controls. Adopted oil and gas storage tank measurement system, hydrocarbon liquids loadout, leak detection and repair, well plugging, and pneumatic controller requirements. Adopted an oil and gas transmission and storage segment methane intensity program. Adopted an annual oil and gas inventory program. Expanded SIP requirements to 50 tpy sources. Aligned gasoline tank truck testing requirements with federal requirements as SIP clean-up.	Sections I. through XX. and Appendices A through F	(see reorganization cross walk)	

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2020	Sept. 23	Adopted requirements for natural gas fired 1,000 horsepower engines. Adopted flowback vessel control requirements and pre- and early-production monitoring requirements. Expanded hydrocarbon liquids loadout requirements to class II disposal well facilities.		Part D, Sections II.; IV.; V.; VI.; Part E, Section I.	Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2020	Dec. 18	Adopted requirements for major source foam manufacturing, boilers, turbines, landfill and biogas fired engines, and wood surface coating.		Part D, Section II.; Part E, Sections II.; IV.; V.	Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2021	Feb. 18	Adopted non-emitting pneumatic controller requirements for new facilities and existing pneumatic controller retrofit requirements for existing facilities.		Part D, Section III.	Regulation 7, Part B (fkna Part D)
2021	July 16	Adopted requirements for metal parts surface coating and major source process heaters.		Part C, Section I.; Part D, Section III.; Part E, Section II.	Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2021	Dec. 17	Adopted SIP revisions to address EPA concerns with the EPA Oil and Gas CTG. Adopted oil and gas combustion device performance testing requirements. Expanded reciprocating compressor rod packing, leak detection and repair, and pneumatic controller requirements at natural gas processing plants. Expanded leak detection and repair, separator, and well maintenance requirements. Adopted pigging and blowdown requirements.		Part D, Sections I., II., III., V., VI.	Regulation 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2022	Dec. 15	Adopted requirements for major source combustion equipment, wood coating, solvent use, bakery operation, digital printing, poultry waste processing, oil stabilization facilities, class II injection well facilities, and industrial waste; included state only provisions as SIP strengthening measures; clarified the applicability of requirements to newly classified ozone nonattainment areas; included requirements for motor vehicle materials and automotive coatings; expanded gasoline tank truck testing requirements.		Part E, Sections I., II., III., VI., VII., and VIII., Part C., Sections I., II., and IV., and Part D, Section II.; Part D, Sections II., ; Part A, Sections I. and II.; Part C, Section I.; and Part B, Section IV.	Regulation 24, Part B (fkna Part B); Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)

The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

These revisions are administrative in nature and, therefore, do not exceed or differ from the requirement of the federal act or rules. Therefore, § 25-7-110.5(5)(a) does not apply.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

(I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.

(II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of greenhouse gas and VOC emissions.

(III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.

(IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.

(V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

II. April 15-17, 2026 (Revisions to Part B, Section VI.)

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

On June 2, 2022, House Bill (HB) 22-1244: Public Protections for Toxic Air Contaminants was signed into law. The law, now primarily codified at Section 25-7-109.5, C.R.S., directed the Commission to adopt emission control regulations for each identified priority toxic air contaminant (PTAC) by April 30, 2026.

On January 17, 2025, the Commission adopted Regulation Number 30 and identified five initial PTACs. In this action, the Commission adopted revisions to Regulation 24, as well as Regulation Number 26 and Regulation Number 30, to establish emissions control regulations for certain stationary sources that emit one or more PTACs.

The emissions control regulations will reduce PTAC emissions, focusing on specific source categories for this initial set of strategies. Specifically, in Regulation Number 24, the adopted emissions control requirements will further reduce emissions of benzene from petroleum liquids refining operations.

Specific Statutory Authority

The State Air Act, specifically Section 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in Section 25-7-102, C.R.S., and that are necessary for the proper implementation and administration of the State Air Act. Section 25-7-109.5(7)(a), C.R.S. required the Commission to, no later than April 30, 2026, adopt emission control

regulations to reduce emissions of each PTAC and prioritize reduction in disproportionately impacted communities with multiple sources of emissions of PTACs.

In determining the emission control regulations, Section 25-7-109.5(7)(b), C.R.S. requires that the Commission consider:

- (I) Any emission control regulations adopted for PTACs in other states or by the federal government;
- (II) The emission levels of a PTAC from different industries and categories of sources, including sources required to have an operating permit pursuant to Section 25-7-114.3, synthetic minor sources, and minor sources;
- (III) The degree of reduction of each PTAC that is achievable and technically and economically feasible, taking into account energy, environmental, and economic impacts and other costs pursuant to the requirements described in Section 25-7-110.8;
- (IV) The ability of emission control regulations to reduce or eliminate the emissions of a PTAC, including non-emitting alternative processes and control technologies; and
- (V) The availability, suitability, and relative efficacy of a less hazardous substitute for a PTAC.

In addition, for new PTAC emission sources, Section 25-7-109.5(7)(c), C.R.S. directs the Commission to adopt emission control regulations that are more stringent than those adopted for existing sources. It also provides that the Commission may adopt an emission threshold below which new sources are not required to comply with the more stringent regulations.

“Emission control regulation” is broadly defined at Section 25-7-103(11), C.R.S. Emission control regulations pertaining to PTACs “may include application of measures, processes, methods, systems, or techniques, including, but not limited to, measures that: (a) [r]educe the volume of, or eliminate emissions of, such pollutants through process changes, emission limitations, control technologies, substitution of materials, or other modifications; (b) [e]nclose systems or processes to eliminate emissions; (c) [c]ollect, capture, or treat such pollutants when released from a process, stack, storage, or fugitive emissions points; (d) [a]re design, equipment, or work practice standards (including requirements for operator training or certification); or (e) [a]re a combination of the provisions of paragraphs (a) to (d)”

In addition to Section 25-7-109.5, C.R.S., Section 25-7-109, C.R.S. provides the Commission authority to adopt emission control regulations for PTACs. Section 25-7-106(6), C.R.S. provides the Commission authority to require sources to establish and maintain reports; install, use, and maintain monitoring equipment or methods; record, monitor, and sample emissions; and provide such other information as the Commission may require.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 24.

PTAC emissions control regulations

The Commission adopted the emissions control regulations for the five initial PTACs based on research and stakeholder outreach conducted by Division staff. Staff developed the emissions control regulations following the direction at Section 25-7-109.5(7)(b), C.R.S. Staff compiled and analyzed emissions data from stationary sources, and the processes that result in emissions. Staff researched emissions control regulations for the five PTAC as adopted in other states and by the federal government, emission levels from different types of source categories, considered the degree of PTAC emissions reductions achievable and technically and economically feasible, considered the ability of emission control regulations to reduce or eliminate PTAC emissions, and considered the availability, suitability, and efficacy of less hazardous substitutes. Staff also held informational meetings with the public and industry groups, as well as source-specific stakeholder meetings throughout the rule development process to ensure adopted provisions effectively reduce PTAC emissions in ways that are technically and economically feasible.

To further reduce emissions of benzene from equipment leaks at petroleum refining operations, the Commission adopted revisions to Regulation Number 24 to revise leak detection and repair requirements at petroleum refineries. Specifically, the Commission modified the requirements for monitoring programs to require recording and, subsequently, repair of leaks with a VOC concentration above 500 ppmv for non-pump equipment and 2,000 ppmv for pumps. As benzene is a component of VOCs, the reduction in VOC emissions that will result from the lower detection limits will result in the reduction of benzene emissions.

The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Incorporation by Reference

The Commission will update regulatory references as needed and as opportunities arrive.

Additional Considerations

The requirement to develop emissions control regulations for PTAC is not a federal rule. However, the Commission recognizes that state air toxic regulations often complement gaps in federal frameworks, and therefore the Commission adopted certain revisions to Regulation Number 24 to achieve additional reductions and compliance oversight of these PTAC sources. The Commission provides the following additional statement, in accordance with § 25-7-110.5(5)(b), CRS.

- (I) *Any federal requirements that are applicable to this situation with a commentary on those requirements;*

The Commission adopted revisions in Regulation Number 24 to modify the definitions of vapor loss and leaks at petroleum refineries and the monitoring programs to require the recording and, subsequently, repair of leaks. These facilities may also be subject to EPA's NSPS J, NSPS Ja, NSPS XX, NSPS GGG, NSPS GGGa, NSPS QQQ, NESHAP CC, NESHAP R and NESHAP UUU. The adopted leak detection limits are more stringent than the ones in the federal rules and do not conflict with the federal requirements.

- (II) *Whether the applicable federal requirements are performance-based or technology-based and whether there is any flexibility in those requirements, and if not, why not;*

The federal rules discussed in (I) are primarily technology-based in that they largely prescribe the use of specific technologies or work practices to comply. There are flexibilities in how these reductions can be achieved. Applicable NESHAP and NSPS are performance-based and provide flexibility to states and sources regarding how to reduce emissions and otherwise meet performance standards.

- (III) *Whether the applicable federal requirements specifically address the issues that are of concern to Colorado and whether data or information that would reasonably reflect Colorado's concern and situation was considered in the federal process that established the federal requirements;*

EPA develops NSPS or NESHAP considering national information and data, not Colorado specific issues or concerns. The above referenced federal requirements concerning petroleum storage are not as stringent as those adopted by the Commission in Regulation Number 24.

- (IV) *Whether the proposed requirement will improve the ability of the regulated community to comply in a more cost-effective way by clarifying confusing or potentially conflicting requirements (within or cross-media), increasing certainty, or preventing or reducing the need for costly retrofit to meet more stringent requirements later;*

The adopted rules will provide certainty to sources, by providing specified technology-based performance standards and clear timing requirements. The adopted rules aid Colorado in achieving emissions control requirements as directed by Section 25-7-109.5(7) and provides regulated entities flexibility to identify and cost-effectively employ emissions control technologies to reduce PTAC emissions in the state.

- (V) *Whether there is a timing issue which might justify changing the time frame for implementation of federal requirements;*

These are state-specific rules that are not addressing federal requirements. Thus, there are no timing issues related to other applicable NESHAP or NSPS.

- (VI) *Whether the proposed requirement will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth;*

The adopted rules will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth as it does not limit sources to specific technology to comply and aligns with regulations seen in federal and other state programs. Additionally, the rules establish reasonable times for meeting PTAC emissions control requirements. The rules afford the ability to affect required emissions controls through alternative compliance measures where needed and establish multiple pathways for compliance.

- (VII) *Whether the proposed requirement establishes or maintains reasonable equity in the requirements for various sources;*

The Commission considered source emissions rates and locations to identify sources of concern for this emission control regulation.

- (VIII) *Whether others would face increased costs if a more stringent rule is not enacted;*

If a more stringent rule is not enacted, public health and welfare and the environment could face increased costs associated with delays in implementation of appropriate PTAC measures.

- (IX) *Whether the proposed requirement includes procedural, reporting, or monitoring requirements that are different from applicable federal requirements and, if so, why and what the “compelling reason” is for different procedural, reporting, or monitoring requirements;*

There are separate requirements to report or monitor PTAC from applicable federal regulations. Federal regulations have different monitoring and reporting thresholds for different types of source categories. In order for Colorado to protect public health and welfare and the environment, state-specific regulations that require more stringent or specific requirements are necessary. The Commission is granted additional authority to adopt PTAC-specific considerations under Section 25-7-109.5 (1)(d), C.R.S.: Emissions of toxic air contaminants should be monitored and controlled to levels that avoid, minimize, or mitigate harm to public health and the environment; and (2)(b) Analyze public health risks from toxic air contaminant emissions from sources based on verified science and reporting of emissions data, including consideration of background levels in the ambient air.

Where applicable, the Commission has not adopted additional procedural, reporting or monitoring requirements where the federal requirements satisfy Colorado specific concerns.

- (X) *Whether demonstrated technology is available to comply with the proposed requirement;*

In aligning with requirements in Section 25-7-109.5(7)(a), C.R.S., the Commission adopted provisions that may require technology that has been demonstrated as available and technically feasible in other states or federal regulations.

- (XI) *Whether the proposed requirement will contribute to the prevention of pollution or address a potential problem and represent a more cost-effective environmental gain;*

The adopted requirements will prevent pollution of identified PTAC, as directed by HB 22-1244. Anticipated reductions in harmful air pollutants are expected to have positive health benefits for the people of Colorado. Cost effectiveness of the regulations are outlined in the rulemaking materials such as the Economic Impact Analysis and various technical support documents.

- (XII) *Whether an alternative rule, including a no-action alternative, would address the required standard.*

The Commission is required to adopt emission control regulations that reduce emissions of each PTAC and prioritize reductions in disproportionately impacted communities with multiple sources of emissions of PTACs. A no-action alternative would not meet the statutory requirements. Alternative rules could address this directive, such as different compliance timelines and emission control requirements. However, the Commission believes that for this initial promulgation of PTAC emission control strategies, the adopted categorical requirements best meet the statutory considerations in Section 25-7-109.5(7)(b), C.R.S.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of PTAC emissions.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.

- (IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.
- (V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

Editor's Notes

History

New rule eff. 06/14/2023.

Rules VI C.2.a, VI C.3, VI C.4.a(ii)-(iii), VI C.4.b(i), VI C.4.b(i)(J), Part C II eff.
06/14/2026.